

Green synthesized silver nanoparticles from *Ocimum sanctum*: A potent inhibitor of biofilm forming ability and efflux pumps in bacteria causing bovine mastitis

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ABSTRACT

Therapeutic management of mastitis faces significant challenges due to multidrug resistance. In the present study multi-drug-resistant (MDR) *Staphylococcus* spp, *Klebsiella pneumoniae* and *Escherichia coli* were isolated from bovine clinical mastitis cases and the phenotypic and genotypic multidrug resistance profiling was carried out. Silver nanoparticles (AgNPs) were biosynthesized using *Ocimum sanctum* leaf extracts and characterized via UV Vis absorption, Fourier Transform Infrared Spectroscopy, Xray Diffraction studies, Energy dispersive spectroscopy and Electron Microscopy. The determined minimum inhibitory concentration and minimum bactericidal concentration of the AgNPs against the recovered MDR isolates were 62.5µg/ml and 125 µg/ml respectively. At a concentration of 50 µg/ml, the AgNPs demonstrated biofilm inhibitory activities of 80.35 % for MDR *E. coli*, 71.29 % for *S. aureus* and 60.18 % for MDR *K. pneumoniae*. Post- treatment observations revealed notable differences in biofilm formation across bacterial isolates. Furthermore, AgNP treatment led to

significant down regulation of expression of the efflux pump genes *acrB*, *acrE*, *acrF* and *emrB* in Gram- negative isolates and *norB* in *Staphylococci* isolates. This research underscores the potential for the development of an ecofriendly antimicrobial alternative in the form of green synthesized silver nanoparticles to combat drug resistance offering potential antibiofilm and efflux pump inhibitory activities.

Keywords: Antimicrobial resistance, clinical mastitis, biofilm, Green synthesis, silver nanoparticles, efflux pump.

1. Introduction

Antimicrobial resistance (AMR) stands as a formidable challenge in global health, with antibiotic misuse being a prime contributor to this looming crisis (1). In dairy cattle, mastitis, a significant economic affliction, has seen escalating antibiotic resistance due to the indiscriminate use of antibiotics. The intrinsic mechanism that contributes to antimicrobial resistance in mastitis pathogens such as *Staphylococcus aureus*, *Escherichia coli* and *Klebsiella pneumoniae* has been correlated with the presence of resistance genes, efflux pump genes and biofilm formation (2-5). However, despite advances in understanding mastitis and its challenges, it remains a daunting issue for both veterinarians and livestock owners underscoring a gap in the therapeutic arsenal. As antimicrobial resistance escalates in pathogens responsible for clinical mastitis, there is an urgent need to seek innovative therapeutic solutions.

In search of innovative solutions, nanotechnology stands out as a promising alternative. Specifically, nanoparticles, typically ranging from one to 100nm, offer unique attributes for combating bacterial resistance mechanisms (6). Silver nanoparticles (AgNPs) could interact with the bacterial cell components and resulted in oxidative stress, permeability and gene expression changes, enzyme inhibition, protein deactivation and electrolyte imbalances resulting in antimicrobial activity (7-9). Compared to the chemical methods in the synthesis of metallic nanoparticles, natural agents, including the microorganisms and plants, offer environment- friendly solutions. The use of plant extracts, notably from *Ocimum sanctum* (Tulsi), an Indian medicinal herb, stands out as a more sustainable method than conventional chemical processes for synthesizing silver nanoparticles (10).

Ocimum sanctum (Tulsi) belonging to the family Lamiaceae is a medicinal herb abundantly found in the entire Indian subcontinent. It is an erect and branched sub-shrub with green or purple leaves. The natural habitat of Tulsi varies from sea level to an altitude of 2000 m. It grows naturally in moist soil all over the globe. (11). Tulsi leaves have medicinal properties, encompassing antimicrobial, antioxidant and anti-inflammatory effects, stemming from their rich biochemical composition (12). Specifically, the carbonyl and hydroxyl groups in Tulsi act as crucial agents in nanoparticle synthesis, reducing and stabilizing the particles. Furthermore, the plant's components ranging from proteins to terpenoids, serve as potent reducing agents in this synthesis (13).

While silver nanoparticles have demonstrated promise in various domains of healthcare, their potential against biofilm formation and efflux pump gene expression in multi-drug- resistant (MDR) strains in veterinary medicine, particularly in the context of mastitis remains largely unexplored. Various researchers have correlated biofilm formation with the rising multidrug resistance in mastitis (14-15) whereas the role of the various efflux pump genes in multi-drug-resistant bacterial isolates in bovine mastitis was not studied extensively. Harnessing the unique properties of biologically synthesized AgNPs, especially those derived from eco-friendly sources like *O. sanctum* leaves, this study offers a pioneering exploration in to their therapeutic efficacy with special reference to antibiofilm and the inhibitory activity against the RND (Resistance Nodulation Division) efflux pumps in the *acrAB* system and the MFS (Major

Facilitator Super family) efflux pump *emrB*. This research aims at synthesizing AgNPs from the leaves of *O. sanctum* to investigate their antimicrobial, anti-biofilm as well as efflux pump inhibition properties against the MDR strains of bacteria (*Staphylococcus aureus*, *S. saprophyticus*, *E. coli* and *Klebsiella pneumoniae*) recovered from clinical cases of bovine mastitis

2. Materials and Methods

2.1. Culture and maintenance of multi-drug- resistant isolates

Antibiotic susceptibility testing of the isolated strains of *S. aureus*, *S. saprophyticus*, *E. coli* and *K. pneumoniae* recovered from bovine clinical mastitis was carried out as per the Clinical and Laboratory Standards Institute guidelines (16-17) using a panel of commercial antibiotic discs (Ceftriaxone-tazobactam, Cefoperazone, Enrofloxacin, Tetracycline, Gentamicin, Amoxicillin-clavulanate and Sulpha-Trimethoprim; HiMedia Laboratories Pvt. Ltd., India). Cefoxitin and oxacillin discs were also included to identify methicillin resistance in the Gram-positive isolates. *E. coli* – ATCC 25922 was used as the negative control.

2.2. Characterization of the MDR isolates

2.2.1. Phenotypic drug resistance profiling

In the current study isolates showing phenotypic resistance to a minimum of the three different classes of antibiotics studied, were considered as MDR isolates after analyzing the antibiotic sensitivity results using the Kirby-Bauer disc diffusion method. The multiple antibiotic resistance index of the isolates was calculated (18) by

$$\text{MAR index for isolates} = \frac{\text{No. of antibiotics resistant to the isolates}}{\text{No. of antibiotics exposed}}.$$

2.2.2. Biofilm forming ability of the recovered multi-drug - resistant isolates

Biofilm forming ability of the recovered multi-drug-resistant isolates was assessed by phenotypic Congo red agar and Microtitre plate assays.

2.2.2.1. Congo red agar (CRA) test

Congo red agar plates were prepared and plates were inoculated with the bacterial isolates and incubated aerobically for 24h at 37°C (19-21).

2.2.2.2. Microtiter plate assay (MTP)

The bacterial isolates were incubated in BHI broth and the grown colonies were diluted in 1:20; 200 µL is poured on to the 96-flat bottom well microtiter plates. Negative control wells contained broth only. The bacterial isolates were classified into non-biofilm producers ($\text{OD} \leq \text{ODc}$), weak biofilm producers ($\text{ODc} < \text{OD} \leq 2\text{xODc}$), moderate biofilm producers ($2\text{ODc} < \text{OD} \leq 4\text{xODc}$), and strong biofilm producers ($4\text{xODc} < \text{OD}$) (22-23).

2.2.3. Identification of the antibiotic- resistance genes

The molecular confirmation of the MDR isolates was done by PCR (24-28). The various antibiotic resistant genes conferring resistance to different antibiotics were identified by end point PCR: beta lactam antibiotics (*blaZ* , *mecA* , *tet M*, *bla SHV*, *bla TEM*, *bla CTX-M*) tetracyclines (*tet A*, *tetB*, *tetC*) and aminoglycosides (*strA/strB*) (29-34). {Supplementary

130 material 1}

131 2.2.4. Identification of the multidrug efflux pumps

132 The total RNA from MDR *Staphylococci*, *E. coli* and *K. pneumoniae* was extracted after lysing
133 the bacterial cell wall using enzymatic lysis buffer (35-36). Complementary DNA (cDNA)
134 synthesis was carried out from total RNA using Revert Aid first strand cDNA synthesis kit
135 (K1622, ThermoScientific, USA) as per manufacturer's protocol. Identification of efflux pump
136 genes in the MDR isolates was done by end point RTPCR targeting the efflux pump genes
137 *norB*, in *Staphylococci* isolates and *acrA*, *acrB*, *acrE*, *acrF*, and *emrB* in *E. coli* and *K.*
138 *pneumoniae* isolates (37-39) {Supplementary material 2}. The amplicons of the efflux pump
139 genes were sequenced and submitted to NCBI GenBank (OR761833, OR761834, OR761835,
140 OR905582)

141 2. 3. Green synthesis of AgNPs

142 Fresh *O. sanctum* leaves were harvested and thoroughly rinsed with deionized water.
143 Approximately 20g of these leaves were finely chopped and mixed with 100mL of deionized
144 water in a 250mL conical flask. This mixture was then heated at 60°C for 20 minutes. Post
145 boiling, the decoction was cooled and subsequently filtered through Whatman paper No. 1,
146 resulting in a transparent yellow broth. This filtrate was stored at 4 °C for ensuing analysis
147 (40).

148 For the green synthesis of AgNPs, 20mL of *O. sanctum* leaf extract was combined with 80mL
149 of 0.1M silver nitrate. This mixture was stirred continuously at 300rpm and 60°C for two hours.
150 The colour changes of the reaction mixture to light brown indicated the formation of silver
151 nanoparticles. These nanoparticles were subsequently purified by washing them three times
152 with methanol and then with nano- pure water . After purification, the AgNPs were airdried at
153 80 °C overnight and stored at 4 °C for subsequent applications.

154 2. 4. Nanoparticle characterization

155 For optical properties, samples were diluted to a concentration of 1mg/mL in ultrapure water
156 and underwent UV-Vis spectroscopy (Thermo Fischer Scientific, USA) . The FTIR
157 spectrometer (Thermo Nicolet iS50) was employed to discern the plant extract's functional
158 groups, focusing on wave numbers between 4000 cm⁻¹ and 500cm⁻¹. The formation,
159 crystalline behavior, and quality of synthesized AgNPs powder were investigated by X-ray
160 diffraction (XRD) spectrum with Cu-K α radiation of 0.15406nm wavelength. The scanning
161 was performed in the region of 2 θ from 10° to 80° at 0.02°/min and the time constant was 2
162 seconds using a Bruker d8 Advance X-ray diffractometer. Scanning electron microscopic
163 analysis was carried out by smearing samples on a small piece of adhesive carbon tape which
164 is fixed on a brass stub and then subjected to gold coating using a sputtering unit (model:
165 JFC1600) for 10 sec at 10mA of current. The gold-coated sample was then placed in the
166 chamber of SEM (Jeol, JSM 6390LA, Japan) and secondary electron/back Scattered electron
167 images were recorded. Elemental analysis was carried out in the scanned area using an EDAX
168 detector (EDAX Make/Model OXFORD XMX N). Transmission electron microscopy was
169 performed on JM 2100 (JEOL, Tokyo, Japan). Samples were prepared by casting the drop of
170 sample solution on carbon coated grids of 200 mesh, then the grid was dried and fixed in the
171 specimen holder. The excess solution was removed by using a piece of soft filter paper. The
172 copper grid was then vacuum-dried.

173 2.5. Antimicrobial studies using AgNPs

Biosynthesized AgNPs were analyzed for their antimicrobial activity against the MDR strains of *S. aureus*, *E. coli* and *K. pneumoniae* isolates. The MIC and MBC of green synthesized AgNPs were carried out to determine the *in-vitro* antibacterial activity (16).

2.5.1. Antibiofilm assay using silver nanoparticles

A 96-well microtiter plate was used to determine the anti-biofilm activity of the AgNPs (41). The percentage of inhibition of biofilm formation was calculated using the equation: biofilm inhibition (in %) = $[1 - (\text{OD } 595 \text{ of cells treated with Ag NPs} / \text{OD } 595 \text{ of the untreated control}) \times 100]$. All the experiments were performed in triplicates.

2.5. 2. Efflux pump Inhibitory activity of silver nanoparticles by Gene expression study

Real-time-quantitative PCR was employed for studying the efflux pump inhibitory activity of Ag NPs. Relative expression of the efflux pump genes *acrB*, *acrE*, *acrF* and *emrB* for MDR *E. coli* isolates was studied in relation to the negative control (*E. coli* – ATCC 25922). In *S. saprophyticus*, relative expression of the efflux pump gene *norB* was studied in relation to the negative control. (Supplementary material 2). The relative expression was analyzed using the formula Fold change = $2^{-\Delta\Delta C_T}$, where $\Delta\Delta C_T = \Delta C_T \text{ treatment} - \Delta C_T \text{ control}$. (42).

2.5.6. Statistical analysis

For estimating the antibiofilm activity, comparison of OD values between isolates, between treated and untreated groups and also the interaction between the isolates and treatment group was done by using two-way ANOVA. Comparison of percentage inhibition between species was done by using one way ANOVA followed by least significant difference test. For the efflux pump inhibitory activity, the results were expressed in fold change as compared to untreated control. Comparison of ΔC_T values between genes, between treated and control groups and also the interaction between genes and treatment group was done by using two-way ANOVA followed by pairwise comparison by using least significant different (LSD) test on least square means

3. Results and Discussion

3.1. Phenotypic MDR profile of the isolates

Four MDR isolates including *S. aureus*, *S. saprophyticus*, *E. coli* and *K. pneumoniae* were selected for the study (table 1). All the four isolates showed resistance to the antibiotics tetracycline and β lactams. Among the two Staphylococci isolates only one was resistant to methicillin phenotypically. The isolates showed MAR index ranging from 0.375 to 0.875. The present findings throw light on the emergence of multidrug resistance in bovine clinical mastitis and the higher MAR index values indicate the increased spreading potential of the MDR isolates which also indicate public health risk (43). Earlier researchers have also indicated the alarming situation of multidrug resistance in bovine clinical pathogens in India and abroad (44-47). A higher resistance pattern may be due to the increased/indiscriminate usage of the above antibiotics in mastitis treatment. Acquired antibiotic resistance is mainly through resistance by mutations, acquisition of resistance genes and biofilm - mediated resistance (48).

218 **Table.1. Multidrug resistance profile of the MDR isolates**
 219

Isolate Id. No	Antibiotics								MDR pattern to different classes of antibiotics	MAR index
	TET	ENR	GE N	β lactam antibiotics						
				C PZ	CFT -TZ	AM XC LV	ME T	SUT		
SA 07	R	R	R		R	R	R	R	TET+ENR+GN+BL + SUT (5)	0.875
SS04	R	R			R				TET+ENR+BL (3)	0.375
EC 04	R	R	R	R	R				TET+ENR+GEN+B L (4)	0.571
KN07	R		R		R	R			TET+GEN+BL (3)	0.571

220
 221 TET- Tetracycline ENR- Enrofloxacin BL- β lactams SUT- Sulpha -trimethoprim GN-
 222 Gentamicin CPZ- Cefoperazone CFT-TZ – Ceftriaxone- Tazobactum AMXCLV- Amoxicillin - Clavulanate
 223 SA01- *S. aureus*, SS01-*S. saprophyticus*, EC-*E. coli* KN -*K. pneumoniae* R-Resistant
 224

225 3.2. Biofilm formation in the isolates

226 3.2.1. Congo red assay

227 The formation and maturation of biofilms follow almost similar pattern in both Gram- positive
 228 and Gram-negative bacteria. All four MDR isolates showed strong biofilm production in CRA
 229 plate indicated by the formation of black colonies with a dry crystalline consistency. The
 230 interpretation of the CRA is based on the color range produced by the bacterial colonies
 231 (Fig.1.). In CRA, Congo red dye directly interacts with polysaccharides, which is produced in
 232 the later stages of biofilm formation and forms coloured complexes (49,20,21). The ability to
 233 form biofilm in vivo is a major virulence factor and thus is responsible for the multidrug
 234 resistance (50).

235 3.2.2. Microtitre plate assay

236 The MTP assay is an important tool for the study of the early stages of biofilm formation
 237 Microtitre plate assay revealed that three of the four MDR isolates showed an OD > 4 OD_c
 238 which indicated strong biofilm production (Fig.2) and the *K. pneumoniae* isolate showed an
 239 OD value in between 2OD and 4OD which denoted an intermediate biofilm production. The
 240 higher percentage of biofilm production in the MDR isolates clearly establishes the role of
 241 biofilm formation in the development of multidrug resistance. A strong correlation between
 242 antibiotic resistance and biofilm production in bacterial isolates was earlier reported (51).



Fig.1.

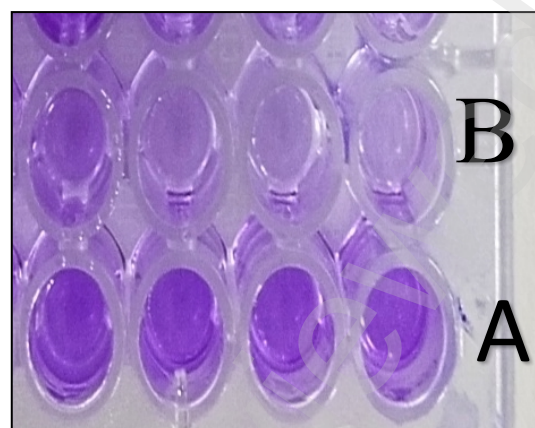


Fig.2.

Fig.1. Formation of black colonies with a dry crystalline consistency indicating strong biofilm production, Fig.2. Crystal violet stained biofilm (A) in the MDR isolates in microtiter plates, Negative control (B).

3.3. Genotypic MDR profile of the isolates

The distribution of the various antibiotic resistance genes and the efflux pump genes in the selected MDR isolates are given in the table.2. Even though only one of the Staphylococci isolates was resistant to methicillin phenotypically, both the isolates harboured the gene *mecA*. All the four isolates harboured any one of the β lactam resistance genes. Among the four three isolates showed the expression of the efflux pump genes also. The genotypic resistance profile clearly indicate the contributing role of the presence of ARGs especially β lactam resistance genes and efflux pump genes towards the development of multidrug resistance. The efflux pump gene *norB* is located in the bacterial chromosome and it confers resistance to the antibiotic class quinolones. The efflux pump gene *norB* was only present in one staphylococci isolate but both the isolates were resistant to quinolones. Hence more extensive study regarding mutations in *gyrA/parC* and resistance conferred by *qnr* gene should be carried out. Hence the multidrug resistance might have occurred due to the synergistic effect of the antibiotic resistance genes, efflux pump genes and biofilm formation

Active efflux of antibiotics can result in cross resistance to unrelated antibiotic class, wide spectrum resistance in bacteria harbouring active efflux pumps along with other mechanisms of resistance and development of mutations in bacteria over expressing efflux pumps (56). There are reports of the presence of the efflux pump genes *acrA* and *acrB* in MDR *K.pneumonie* isolates and the association between the efflux pump *acrAB* and co- resistance of β lactams, quinolones and tetracyclines in *K. pneumoniae* isolates (52-54).

Table 2. Genotypic resistance profile and biofilm formation of the MDR isolates

Isolate ID	Biofilm formation		Gene pattern	
	CRA	MTP OD $\pm$ SE	ARGs	Efflux pump genes
SA 07	Strong biofilm producer	4xODc < OD 0.21 $\pm$ 0.02	<i>mecA</i> , <i>strA/strB</i>	-
SS04	Strong biofilm producer	4xODc < OD 0.21 $\pm$ 0.006	<i>mecA</i> , <i>blaZ</i> , <i>strA/strB</i>	<i>norB</i>

EC 04	Strong biofilm producer	$4 \times \text{ODc} < \text{OD}$ 0.324 ± 0.01	<i>bla</i> TEM,	<i>acrB, acrE, acrF, emrB</i>
KN07	Strong biofilm producer	$2 \text{ODc} < \text{OD} \leq 4 \times \text{ODc}$ 0.144 ± 0.006	<i>bla</i> CTX-M, <i>tetA, tetB</i>	<i>acrB, acrE, acrF, emrB</i>

SA01- *S. aureus*, SS01-*S. saprophyticus*, EC04-*E. coli*, KN 07 -*K. pneumoniae*, CRA- Congo red test, MTP- Microtitre Plate assay, ARGs- Antibiotic resistance genes, OD-Optical density, OD_c . Optical density of the control well- (0.052)

The selected MDR isolates were used to study the antimicrobial, antibiofilm and efflux pump inhibitory activity of the synthesized silver nanoparticles

3.4. Synthesis and characterization of silver nanoparticles

In the present study the aqueous extract of *O. sanctum* (Fig.3 a) was used for the green synthesis of silver nanoparticles. Subsequently, these nanoparticles were subjected to characterization to elucidate their optical attributes, ascertain the role of plant metabolites capping and stabilization and determine their morphology, crystallinity and elemental composition. The study utilized the aqueous extract of *O. sanctum* for the eco-friendly synthesis of silver nanoparticles. The aqueous leaf extract facilitated the reduction of silver nitrate, transforming silver ions into their elemental nano-sized form. The emergence of brown precipitate setting at the base of the beaker served as indicative marker for nanoparticle formation (Fig 3b). This reductive capacity could likely be attributed to the abundant antioxidants present within the *O. sanctum* extract (55).

3.4.1. UV Visible spectra of the synthesized silver nanoparticles

Absorption spectra for the reaction mixture revealed a characteristic peak centered between 400 and 450 nm, indicative of the presence of silver nanoparticles. In Fig.3c UV visible spectra of the synthesized silver nanoparticles is shown. The UV Vis spectral analysis of the reaction mixture showed an absorbance peak which is representative of the bio reduction of silver ions into nanoparticles mediated by the *O.sanctum* leaf extract. The unique optical properties are attributed to silver nanoparticles intrinsic ability to selectively interact with certain light wavelengths. The absorbance phenomenon could be traced back to the surface plasmon resonance of silver nanoparticles, which entails a coordinated oscillation of the free electrons in the conduction band, resonating with the incident light wave. This observation aligns with the findings previously reported (56).

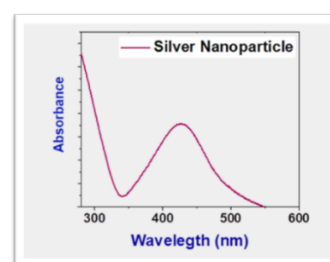


Fig 3 a **Fig. 3b** **Fig. 3c**
Fig 3a. *Ocimum sanctum* leaves, Fig. 3b. Green synthesized silver nanoparticles, Fig.3c UV Visible spectra of the synthesized silver nanoparticles which depict the absorbance peak centered between 400 and 450 nm indicative of silver nanoparticles.

3.4.2. Fourier Transform InfraRed Spectroscopy analysis

FTIR analysis was employed to elucidate the potential biomolecular interactions between the different functional groups in the leaf extract and the silver nanoparticles. Distinctive absorption peaks were identified at 3434 cm^{-1} , 2964 cm^{-1} , 2922 cm^{-1} , 2853 cm^{-1} , 1724 cm^{-1} , 1599 cm^{-1} which corresponded to various functional groups from *O. sanctum* extracts attached to the synthesized silver nanoparticles which are depicted in Fig. 4. These peaks were further analysed against established IR spectra references to ascertain the associated functional groups. FTIR measurements of biosynthesized silver nanoparticles deciphered the possible interactions between various functional groups inherent to the *O. sanctum* leaf extract and the silver nanoparticles. Clear absorption peaks observed points at a protective layer of proteins and metabolites with functional moieties characteristic of alcohols, alkanes and aldehydes enveloping the nanoparticles. The absorption peaks at 3434 cm^{-1} , represent the O-H stretch typical of alcohols. The peaks at 2964 cm^{-1} , 2922 cm^{-1} , 2853 cm^{-1} can be attributed to CH stretching found in alkanes. Furthermore, the peaks 1724 cm^{-1} , 1599 cm^{-1} align with C=O stretch in aldehydes as elucidated by standard IR spectra references. These molecular interactions mirror patterns earlier highlighted (57-58). This analysis strongly supports the capping of the synthesized silver nanoparticles by the metabolites in the leaf extract and underscore the versatility of botanical constituents in nanoparticle synthesis and stabilization.

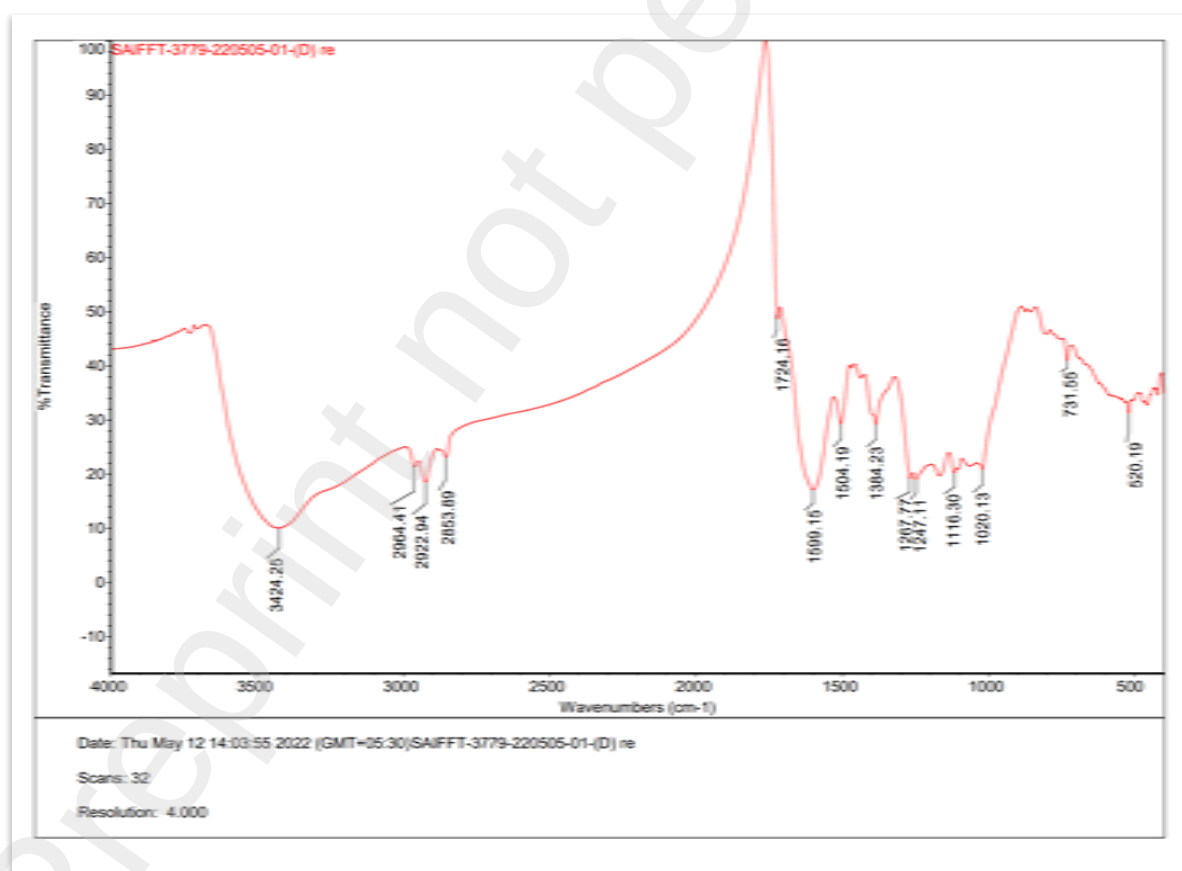


Fig 4. FTIR pattern of biosynthesized silver nanoparticles indicates sharp absorption peaks located at 3434 cm^{-1} , 2964 cm^{-1} , 2922 cm^{-1} , 2853 cm^{-1} , 1724 cm^{-1} , 1599 cm^{-1} which correspond to various functional groups from *O. sanctum* extracts attached to the synthesized silver nanoparticles

3.4.3. Xray Diffraction pattern

X-ray Diffraction (XRD) patterns underscored the crystalline nature of the synthesized silver nanoparticles, evident from the pronounced peaks. Specifically, the distinct peaks were observed at 77.513 °, 64.596 °, 46.393 °, 44.408 °, 38.211 °, 36.170 °, 32.336 ° and 27.954 ° characteristic of silver nanoparticles as shown in Fig 5. These diffraction values were benchmarked against the standard diffraction profiles from the Joint Committee on Powder diffraction standards (JCPDS). X-ray interactions with crystalline structures result in distinct diffraction patterns that offer insights into crystal's inherent physiochemical attributes. In the context of our study, the X-ray diffraction assessment revealed dominant peaks at 77.513 °, 64.596 °, 44.408 ° and 38.211 °. These peaks are characteristic of the face-centered cubic structure of the synthesized silver nanoparticles. More specifically, these peaks correspond to the 311, 220, 200 and 111 reflection planes, respectively. Such diffraction patterns and their association with specific crystal planes in silver nanoparticles have been previously delineated (57). The consistency in these findings underscores the robust nature of the nanoparticle synthesis process and the distinct crystalline characteristics inherent to silver nanoparticles.

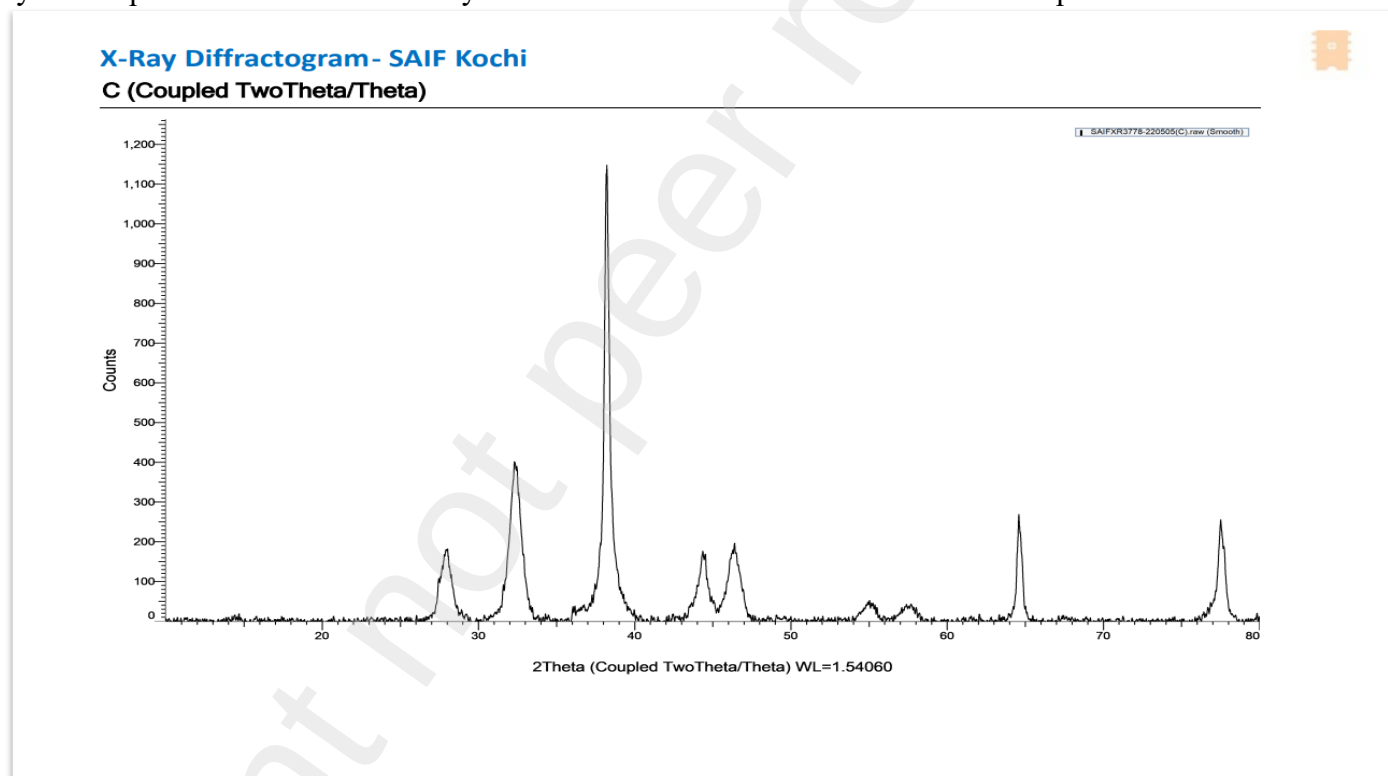


Fig 5. XRD pattern of biosynthesised silver nanoparticles indicates distinct peaks at 77.513 °, 64.596 °, 46.393 °, 44.408 °, 38.211 °, 36.170 °, 32.336 ° and 27.954 ° characteristic of silver nanoparticles

3.4.4. Scanning Electron Microscopy and EDAX

The Scanning Electron Microscopy images depicted the morphology and distribution of synthesized silver nanoparticles, notably revealing agglomerates of various particles as visualized in Fig. 6 at 50 µm resolutions. Energy dispersive X-ray spectroscopy (EDAX) analysis illuminated the elemental composition of these nanoparticles. A pronounced signal attributed to silver was evident at 3 keV, stemming from surface plasmon resonance. Concurrently, weaker signals alluded to the presence of organic components, likely derived from enzymes and proteins. Notably, the EDAX quantitative analysis registered a predominance of elemental silver, accounting for approximately 58.75 (wt. %) as shown in Fig

7. Utilizing a combined approach with SEM and EDAX, the morphological attributes and elemental composition of the synthesized silver nanoparticles were determined. SEM imaging revealed the distinct crystallinity of these nanoparticles. The EDAX provided an analytical insight in to their elemental composition. The nanoparticles predominantly contained silver (58.75%), accompanied by other elements such as Carbon (18.97%), Oxygen (13.82%), Chlorine (7.15%) and Copper (1.31%). The pronounced peak observed at 3 k eV in the EDAX spectrum was indicative of surface plasmon resonance of silver nanoparticles, confirming the elemental silver's dominance. Conversely, faint signals observed might be attributed to residual biomolecules or other compounds from the synthesis process. The quantitative results of EDAX spectra showed a yield of 58.75(wt.%) of elemental silver.

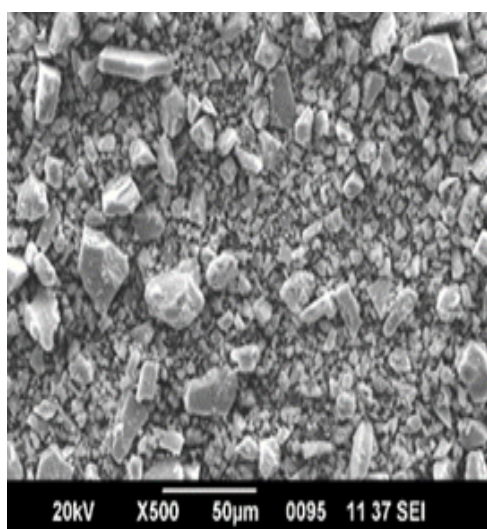


Fig.6

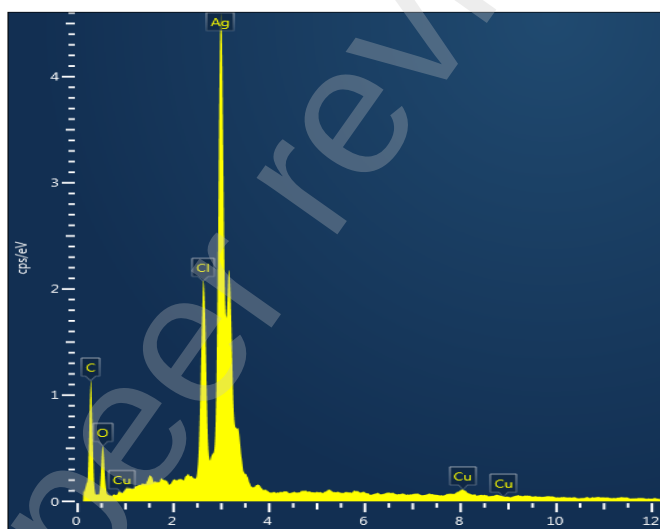


Fig.7.

Fig 6 SEM images of the silver nanoparticles indicates the crystalline nature, shape of the crystals and the even distribution. Fig 7. EDAX quantitative analysis registered a predominance of elemental silver, accounting for approximately 58.75 (wt. %)

3.4.5. Transmission Electron Microscopy analysis

The TEM images of the prepared silver nanoparticles at 10 nm scale as shown in Fig.8 revealed that Ag nanoparticles were spherical in shape with particles in size range within 6.29-10.61 nm with mean diameter $\pm$ SD of 8.287 ± 1.389 nm. It was also observed that silver nanoparticles were evenly distributed in the sample. Transmission electron microscopy further highlighted the nanoscale morphology of silver nanoparticles. The particles were predominantly spherical with a diameter range of 6.29-10.61 nm with mean diameter of 8.287 nm and distributed uniformly throughout the sample. This morphological observation especially regarding the spherical shape, aligns with earlier findings which documented similar morphological characteristics in their study of silver nanoparticles (59).

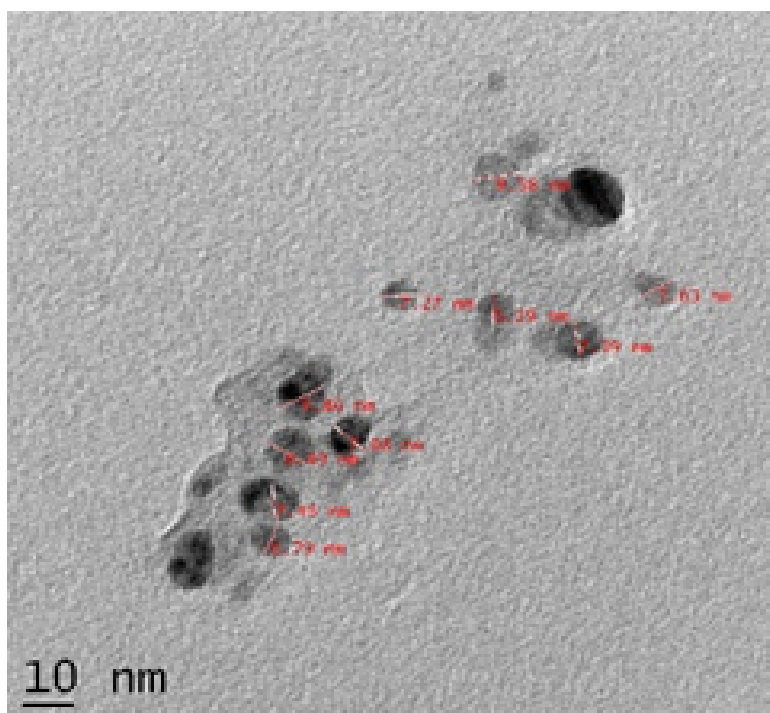


Fig 8. TEM image of the silver nanoparticles indicating Ag nanoparticles were spherical in shape with particles in size range within 6.29-10.61 nm with mean diameter $\pm$ SD of 8.287 ± 1.389 nm

3.5. Antimicrobial efficacy of biosynthesized silver nanoparticles

The antimicrobial efficacy of biosynthesized silver nanoparticles was evaluated from the MIC. The minimum inhibitory concentration against the MDR isolates was observed as 62.5 $\mu\text{g/ml}$. MBC of the silver nanoparticles against the MDR *S. aureus*, *E. coli* and *K. pneumoniae* isolates were evaluated based on the absence of growth in the media as 125 $\mu\text{g/ml}$ which is similar to findings by earlier researchers who also reported an MIC of 125 $\mu\text{g/ml}$ to *K. pneumoniae* isolates (60). Our data showed that silver nanoparticles have potent antibacterial activity against MDR isolates. Previous studies (61-62) have documented the antibacterial activity of green synthesized silver nanoparticles against *E. coli*, *K. pneumoniae* and *S. aureus*. However, the underlying mechanisms fostering this antimicrobial activity remains a subject of debate and research. Silver is recognized for its antimicrobial attributes, is believed to confer enhanced potency to its nanoparticulate form due to the latter's small dimensions and extensive surface area. This allows the silver nanoparticles to efficiently adhere to the cell walls and penetrate into the bacteria easily. The formation of free radicals, inactivation of sulfur-containing proteins and phosphorus-containing compounds like DNA in the bacteria, and the production of reactive oxygen species that induce oxidative stress were the various reasons reported for the antibacterial efficacy of silver nanoparticles. Furthermore, silver nanoparticles are believed to catalyze the release of reactive oxygen species, including oxidative stress in bacterial cells, further contributing to their bactericidal action (41).

3.6. Antibiofilm activity of the synthesized silver nanoparticles

Antibiofilm activity of the synthesized silver nanoparticles against the MDR *S. aureus*, *E. coli* and *K. pneumoniae* isolates were evaluated using the biofilm inhibition percentage of the silver nanoparticles. Three concentrations of Ag nanoparticles were used against the tested microorganisms; 50 $\mu\text{g/ml}$, 100 $\mu\text{g/ml}$, 200 $\mu\text{g/ml}$. Biofilm inhibition was observed at 50 $\mu\text{g/ml}$ in comparison with control. The OD570 value of *S. aureus* was 0.210 ± 0.025 without AgNPs and with AgNPs had an OD value 0.058 ± 0.004 ; Hence, the green synthesized AgNPs

triggered 71.29 % drop in the biofilm as compared to the control. The OD570 value of *E. coli* was 0.324 ± 0.016 without AgNPs and with AgNPs had an OD value 0.063 ± 0.007 ; Hence, the green synthesized AgNPs triggered 80.35 % drop in the biofilm as compared to the control. The OD570 value of *K. pneumoniae* was 0.144 ± 0.006 without AgNPs and with AgNPs had an OD value 0.057 ± 0.005 . Hence, the green synthesized AgNPs triggered 60.18 % drop in the biofilm as compared to the control as shown in Fig .9. Significant F-value between treated and untreated shows that there exists significant difference in OD values between treated and untreated groups. The F-value of the comparison of percentage inhibition between the isolates was found to be significant at 0.05 level as the p-value is less than 0.05. Hence, pair wise comparison between the isolates was done by using the LSD test. Results showed that percentage inhibition was significantly higher in *E. coli* compared to *K. pneumoniae*.

One of the main objectives of the study was to explore the potential of biosynthesized silver nanoparticle in thwarting biofilm formation, a primary factor underpinning multidrug resistance in bacterial pathogens. Our investigations unearthed that the silver nanoparticles manifested substantial antibiofilm activity, inhibiting 80.35 % of biofilm in MDR *E. coli* and 60.18% for MDR *K. pneumoniae* isolates. Consistent with our findings, earlier research work has reported the antibiofilm activity of silver nanoparticles against *S. aureus* and *E. coli* (41). The inhibitory action of silver nanoparticles against biofilms is believed to be multifaceted. Specifically, these nanoparticles prevent bacterial adhesion to surfaces, which is a pivotal step in biofilm formation. Additionally, they disrupt quorum sensing, a bacterial communication mechanism vital for biofilm maturation and maintenance (62-63). Such attributes underscore the potential of silver nanoparticle as promising agents in combating biofilm-associated challenges in clinical and environmental settings.

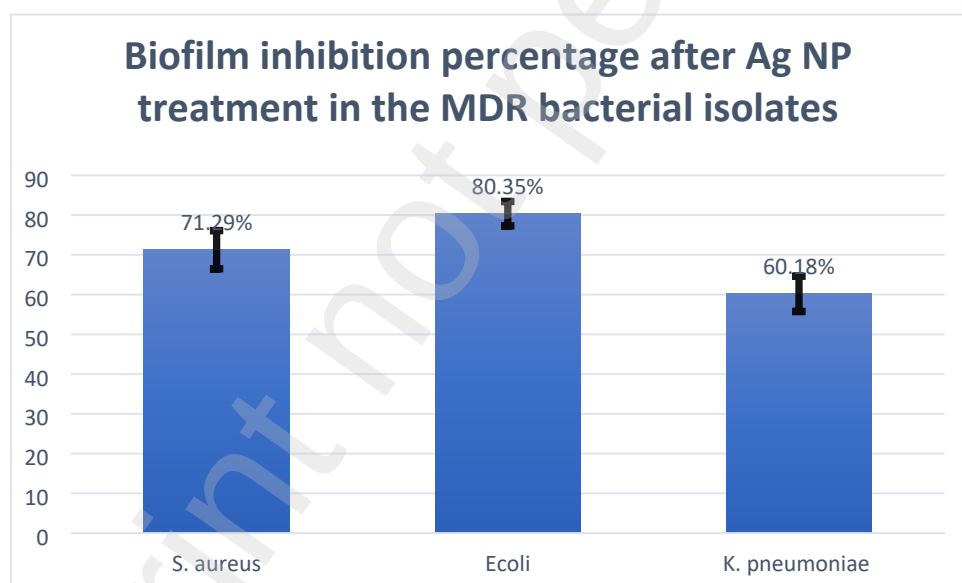


Fig 9 X axis-Biofilm inhibition percentage after Ag NP treatment in the MDR bacterial isolates. *S. aureus* - 71.29 % drop in the biofilm production, *E. coli* - 80.35 % drop in the biofilm production, *K. pneumoniae*- 60.18 % drop in the biofilm production. Y axis- Biofilm inhibition percentage

3.7. Efflux pump inhibitory activity of the synthesized silver nanoparticles

The influence of biosynthesized AgNPs on efflux- pump mediated resistance was explored by assessing the mRNA expression levels of pivotal efflux- related genes, including *acrB*, *acrE*, *acrF*, *emrB* and *norB*. Quantitative real- time PCR (qRT-PCR) results demonstrated that the mRNA levels of efflux-related genes were significantly downregulated post AgNPs treatment.

This reduction when compared to the reference housekeeping gene(*rp SL* as control) accentuates the potential of AgNPs in attenuating efflux pump mechanisms at the transcription level as shown in Fig.10.. Significant F-value between treated and untreated shows that there exists significant difference in ΔC_T values between treated and untreated groups which is depicted in Table.5

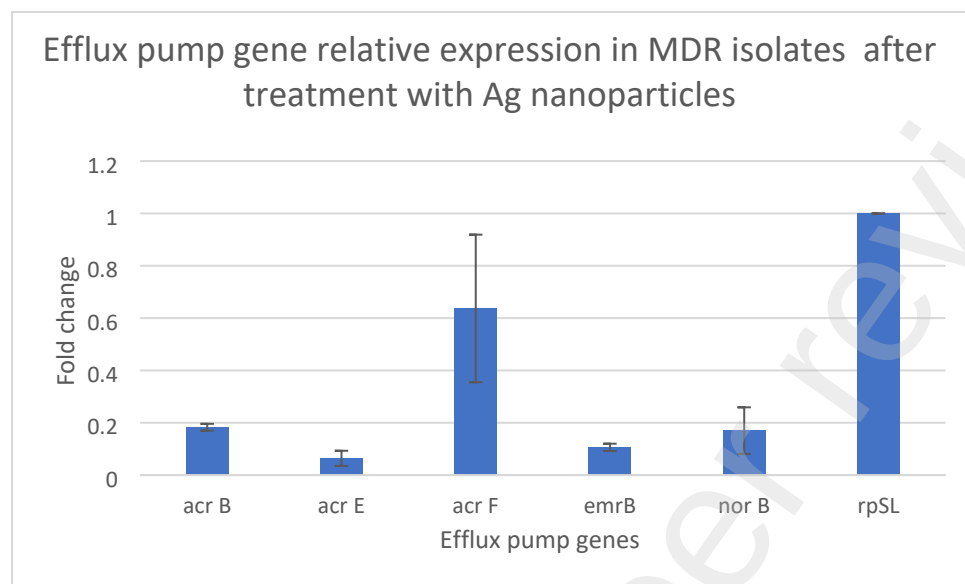


Fig.10. Efflux pump gene relative expression in MDR strains is represented as a fold change between the efflux pump genes and the internal control gene *rpSL* after treatment with subMIC concentration of Ag NPs. Data are mean $\pm$ SE of three independent experiments.

Table. 5. Relative expression of the efflux pump genes in MDR isolates

Genes	ΔC_T control	ΔC_T treatment	$\Delta \Delta C_T$ (ΔC_T treatment – ΔC_T control)	Fold of expression (Treatment)
<i>acr B</i>	4.73 $\pm$ 0.101	7.19 $\pm$ 0.003	2.47 $\pm$ 0.10 ^{ab}	0.183 $\pm$ 0.013
<i>acr E</i>	5.33 $\pm$ 0.395	9.80 $\pm$ 0.546	4.47 $\pm$ 0.71 ^a	0.064 $\pm$ 0.029
<i>acrF</i>	5.90 $\pm$ 0.808	7.18 $\pm$ 0.048	1.29 $\pm$ 0.84 ^b	0.637 $\pm$ 0.282
<i>emrB</i>	1.72 $\pm$ 0.227	5.02 $\pm$ 0.064	3.26 $\pm$ 0.20 ^{ab}	0.106 $\pm$ 0.014
<i>nor B</i>	4.37 $\pm$ 0.442	7.37 $\pm$ 0.700	3.00 $\pm$ 0.82 ^{ab}	0.170 $\pm$ 0.089
F-value (P-value)			3.683* (0.032)	

*** Significant at 0.05 level; ns non-significant**

Means having different letter as superscript differ significantly within a column. Significant F-value between treated and untreated shows that there exists significant difference in ΔC_T values between treated and untreated groups.

In this study, significant emphasis was placed on evaluating the inhibitory activity of synthesized silver nanoparticles on bacterial efflux pumps, crucial mechanism underlying multidrug resistance. Among the Gram-negative bacteria, the most studied efflux pumps are

the RND efflux pumps and the studied genes coding the RND efflux pumps include the *acrA*, *acrB*, *acrE* and *acrF* and they are good targets for inhibition as they confer multidrug resistance. The genes *emrA* and *emrB* coding for the MFS efflux pumps in Gram- negative bacteria were also included in the study.

While extensive research exists on efflux pumps, limited literature addresses the impact of the silver nanoparticles on key efflux systems like the tripartite RND efflux pumps *acrAB-TolC*, *acrEF-TolC* and the tripartite MFS efflux pump *emrAB-TolC* in Gram- negative bacteria and *norB*, one of the most studied MFS efflux pumps of *S. aureus*.

Our findings revealed a marked decrease in the gene expression associated with these efflux pumps following treatment with silver nanoparticles. The mechanism underpinning this inhibitory activity can be attributed to two primary pathways. One involves the competitive inhibition of antibiotics through the direct binding of silver nanoparticles to the efflux pump's active sites. The other is rooted in the alteration of efflux kinetics. Such inhibitory effects might arise from the generation of reactive oxygen species (ROS) by the nanoparticles or their direct interaction with transcriptional elements, ultimately leading to a suppression in the transcription of efflux pump genes (64). These findings pave the way for a deeper understanding of nanoparticle bacterial interactions and their implications in combating antibiotic resistance.

4. Conclusion

In this investigation, we isolated and characterized MDR *S. aureus*, *S. saprophyticus*, *E. coli* and *K. pneumoniae* from bovine clinical mastitis. The phenotypic and genotypic multidrug resistance pattern and biofilm- forming ability of the isolates were identified. Without doubt we could assess, the contributing role of ARGs, biofilm formation and the active efflux of antibiotics towards multidrug resistance. We demonstrated the synthesis of silver nanoparticles using *O. sanctum* leaf extract as a reducing and capping agent. These biogenically synthesised nanoparticles exhibited notable antimicrobial, antibiofilm and efflux-pump inhibitory activity. Intriguingly, our results also underscored the potential of these nanoparticles to notably diminish the expression levels of efflux pump genes in multidrug resistant bacterial isolates associated with bovine clinical mastitis. It's pivotal to highlight that while our findings provide compelling evidence at the transcription level, complementary proteomic studies are essential to corroborate these gene expression patterns. The highlighted efflux pump inhibitory function offers a promising avenue for combating antibiotic resistance particularly in the context of bovine mastitis treatment. Nevertheless, a more comprehensive array of *in vitro* and *in vivo* analyses is imperative to delineate the full spectrum of these nanoparticles' therapeutic potential.

Our findings illuminate the potential of green synthesized silver nanoparticles as powerful tools, demonstrating remarkable antibacterial, antibiofilm and efflux pump inhibitory activity against MDR *Staphylococci*, *K. pneumoniae* and *E. coli*. This research not only bolsters silver nanoparticles as viable alternative to conventional antibiotics but also offers a sustainable pathway for addressing the mounting concern of antibiotic- resistant infections in veterinary medicine.

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517 **Conflicts of interests**

518 The authors declare that they have no competing interests.

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Data availability

All the images given in the article are obtained based on experimental data. None of the images were reproduced from other sources. On request, the experimental data related to experiments can be provided.

Ethical approval

Not applicable

6. References

1. World Health Organization. Antimicrobial resistance: Global report on surveillance. 2014. Geneva, Switzerland: WHO; 2014.
2. Banerjee, A., Batabyal, K., Singh, A.D., Joardar, S.N., Dey, S., Isore, D.P., Sar, T.K., Dutta, T.K., Bandyopadhyay, S., Samanta, I. 2020. Multi-drug resistant, biofilm-producing high-risk clonal lineage of *Klebsiella* in companion and household animals. *Lett. Appl Microbiol.* 71(6). :80-587.
3. Cheng, W.N. and Han, S.G., 2020. Bovine mastitis: risk factors, therapeutic strategies, and alternative treatments-A review. *Asian-Australasian J. Anim. Sci.* 33(11),1699-1713.
4. Lee, A.H., Porto, W.F., de Faria Jr, C., Dias, S.C., Alencar, S.A., Pickard, D.J., Hancock, R.E. and Franco, O.L. 2021. Genomic insights into the diversity, virulence and resistance of *Klebsiella pneumoniae* extensively drug resistant clinical isolates. *Microbial Genomics.* 7(8). DOI 10.1099/mgen.0.000613.
5. Pascu, C., Herman, V., Iancu, I. and Costinar, L. 2022. Etiology of Mastitis and Antimicrobial Resistance in Dairy Cattle Farms in the Western Part of Romania. *Antibiotics.* 11(1): 57.
6. Medda, S., Hajra, A., Dey, U., Bose, P. and Mondal, N.K. 2015. Biosynthesis of silver nanoparticles from Aloe vera leaf extract and antifungal activity against *Rhizopus* sp. and *Aspergillus* sp. *Appl. Nanosci.* 5(7) : 875-880.
7. Tamayo, L.A., Zapata, P.A., Vejar, N.D., Azócar, M.I., Gulppi, M.A., Zhou, X.,Thompson, G.E., Rabagliati, F.M. and Páez, M.A. 2014. Release of silver and copper nanoparticles from polyethylene nanocomposites and their penetration into *Listeria monocytogenes*. *Mater. Sci. Eng*, 40, 24-31.
8. Wang, C., Zhao, M., Li, J., Yu, J., Sun, S., Ge, S., Guo, X., Xie, F., Jiang, B., Wujcik, E.K. and Huang, Y. 2017. Silver nanoparticles/graphene oxide decorated carbon fiber synergistic reinforcement in epoxy-based composites. *Polymer*, 131, 263-271.
9. Tailor G, Yadav BL, Chaudhary J, Joshi M, Suvalka C.2020. Green synthesis of silver nanoparticles using *Ocimum canum* and their anti-bacterial activity. *Biochem Biophys Rep.* 2020;24:100848. Published 2020 Dec 2. doi:10.1016/j.bbrep.2020.100848

10. Selvam, K., Sudhakar, C., Govarthanan, M., Thiagarajan, P., Sengottaiyan, A., Senthilkumar, B. and Selvankumar, T. 2017. Eco-friendly biosynthesis and characterization of silver nanoparticles using *Tinospora cordifolia* (Thunb.) Miers and evaluate its antibacterial, antioxidant potential. *J. Radiation Res. Applied Sci.* 10(1):6-12.
11. Cohen, M. M. 2014. Tulsi - *Ocimum sanctum*: A herb for all reasons. *J Ayurveda Integr Med.* 5(4):251-259. doi:10.4103/0975-9476.146554
12. Mohamad, N. A. N., Arham, N. A., Jai, J., and Hadi, A. 2014. Plant Extract as reducing Agent in Synthesis of Metallic Nanoparticles. *Adv. Materials. Res.* 832 :350. doi: 10.4028/www.scientific.net/AMR.
13. Pattanayak, P., Behera, P., Das, D., & Panda, S. K. 2010. *Ocimum sanctum* Linn. A reservoir plant for therapeutic applications: An overview. *Phcog. Rev.* 4(7), 95–105. <https://doi.org/10.4103/0973-7847.65323>
14. Francisco, M.S., Rossi, C.C., Brito, M.A.V.P., Laport, M.S., Barros, E.M. and Giambiagi-deMarval, M. 2021. Characterization of biofilms and antimicrobial resistance of coagulase-negative *Staphylococcus* species involved with subclinical mastitis. *J. Dairy Res.* 88(2):179-184.
15. Quinn, P., Markey, B., Carter, M. and Carter, G. R. 2013. *Clinical Veterinary Microbiology.* (2nd Ed.) Mosby, St. Louis: 514p.
16. CLSI [Clinical and Laboratory Standards Institute]. 2019. M100. Performance Standards for Antimicrobial Susceptibility Testing (29th Ed). Clinical and Laboratory Standards Institute, Wayne, Pennsylvania, USA, 260p
17. Bauer, A.W. 1966. Antibiotic susceptibility testing by a standardized single disc method. *Am J clin pathol.* 45 : 49-158.
18. Krumperman, P. H. 1983. Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of fecal contamination of foods. *Appl. Environ Microbiol.* 46(1), 165-70. doi: 10.1128/aem.46.1.165-170.1983. PMID: 6351743; PMCID: PMC239283.
19. Freeman, D.J., Falkiner, F.R. and Keane, C.T. 1989. New method for detecting slime production by coagulase negative staphylococci. *J. Clin Pathol.* 42(8):872-874.
20. Darwish, S.F. and Asfour, H.A. 2013. Investigation of biofilm forming ability in *Staphylococci* causing bovine mastitis using phenotypic and genotypic assays. *The Scientific World J.* 2013:378492. doi: 10.1155/2013/378492.
21. Aricola CR, Compoccia D, Baldassarri L, Donati ME, Pirini V, Gamberini S, Montanaro L. 2005. Detection of biofilm formation in *Staphylococcus epidermidis* from implant infections. Comparison of a PCR method that recognizes the presence of *ica* genes with two classic phenotypic methods. *J. Biomed. Mater. Res.* 76 (2): 425- 4230
22. Christensen, G. D., Simpson, W. A., Younger, J. J., Baddour, L. M., Barrett, F. F., Melton, D. M., & Beachey, E. H. 1985. Adherence of coagulase-negative staphylococci to plastic tissue culture plates: a quantitative model for the adherence of staphylococci medical devices. *J. clin. Microbiol.* 22(6):996–1006. <https://doi.org/10.1128/jcm.22.6.996-1006.1985>
23. Stepanović, S., Vuković, D., Hola, V., Bonaventura, G.D., Djukić, S., Ćirković, I. and Ruzicka, F. 2007. Quantification of biofilm in microtiter plates: overview of testing conditions and practical recommendations for assessment of biofilm production by staphylococci. *Apmis*, 115(8) :891-899.
24. McClure, J.A., Conly, J.M., Lau, V., Elsayed, S., Louie, T., Hutchins, W. and Zhang, K. 2006. Novel multiplex PCR assay for detection of the staphylococcal virulence marker Panton-Valentine leukocidin genes and simultaneous discrimination of methicillin-

- susceptible from-resistant staphylococci. *J. Clin. Microbiol.* **44**(3) :1141-1144.
25. Sasaki, T., Tsubakishita, S., Tanaka, Y., Sakusabe, A., Ohtsuka, M., Hirotaki, S., Kawakami, T., Fukata, T. and Hiramatsu, K. 2010. Multiplex-PCR method for species identification of coagulase-positive staphylococci. *J. Clin. Microbiol.* **48**(3) :765-769.
 26. Okolie, C.E., Wooldridge, K.G., Turner, D.P., Cockayne, A. and James, R. 2015. Development of a heptaplex PCR assay for identification of *Staphylococcus aureus* and CoNS with simultaneous detection of virulence and antibiotic resistance genes. *BMC microbiology*. **15**(1): 1-7.
 27. Bej, A.K., Dicesare, J.L., Haff, L. and Atlas, R.M. 1991. Detection of *Escherichia coli* and *Shigella* spp. in water by using the polymerase chain reaction and gene probes for uid. *Appl. Environ. Microbiol.* **57**(4) :1013-1017.
 28. Singh, K., Chandra, M., Kaur, G., Chachra, D and Gupta, D. 2018. Prevalence and Antibiotic Resistance Pattern among the Mastitis Causing Microorganisms. *Open J. Vet. Med.* **08** :54-64. 10.4236/ojvm.2018.84007.
 29. Schnellmann, C., Gerber, V., Rossano, A., Jaquier, V., Panchaud, Y., Doherr, M. G., Thomann, A., Straub, R and Perreten, V. 2006. Presence of new *mecA* and *mph*(C) variants conferring antibiotic resistance in *Staphylococcus* spp. isolated from the skin of horses before and after clinic admission. *J. Clin. Microbiol.* **44**(12), 4444–4454. <https://doi.org/10.1128/JCM.00868-06>.
 30. Alipour, F., Ahmadi, M. and Javadi, S. 2014. Evaluation of different methods to detect methicillin resistance in *Staphylococcus aureus* (MRSA). *J. Infect. Public Health.* **7**(3), 186-191.
 31. Yazdi, M.K.S., Fallah, J., Dalal, M.M.S., Akhi, M.T., Azarsa, M., Shamkani, F., Barhaghi, M.H.S., Aghamirzaei, H.M., Sabbaghi, A. and Mohtasab, T.P.A. 2012. Survey of TEM Type Extended-Spectrum β -Lactamases) ESBL (in Clinical Isolates of *Escherichia Coli* by Phenotypic and Genotypic Methods. *Med. J. Tabriz Univ. Med. Sci.* **34**(1), 56-62.
 32. Ahmed, A.M., Nakano, H. and Shimamoto, T. 2004. The first characterization of extended-spectrum β -lactamase-producing *Salmonella*. *Japan. J. Antimicrob. Chemotherapy.* **54**(1), 283-284.
 33. Mazurek, J., Pusz, P.A.W.E.L., Bok, E., Stosik, M and Baldy-Chudzik, K. 2013. The phenotypic and genotypic characteristics of antibiotic resistance in *Escherichia coli* populations isolated from farm animals with different exposure to antimicrobial agents. *Pol. J. Microbiol.* **62**(2), 173-79.
 34. Kozak, G.K., Boerlin, P., Janecko, N., Reid-Smith, R.J. and Jardine, C. 2009. Antimicrobial resistance in *Escherichia coli* isolates from swine and wild small mammals in the proximity of swine farms and in natural environments in Ontario, Canada. *Appl Environ Microbiol.* **75**(3), 559-566.
 35. Villa-Rodríguez, E., Ibarra-Gómez, C. and de Los Santos-Villalobos, S. 2018. Extraction of high-quality RNA from *Bacillus subtilis* with a lysozyme pre-treatment followed by the Trizol method. *Journal of Microbiological Methods*, **147**, 14-16.
 36. Regan, J.F.; Furtado, M.R.; Brevnov, M.G. and Jordan, J.A. 2012. A sample extraction method for faster, more sensitive PCR-based detection of pathogens in blood culture. *J. Mol. Diagn* **14**(2), 120-129. 10.1016/j.jmoldx.2011.10.001 S1525- 1578(11)00311-4
 37. Swick, M. C, Morgan-Linnell, S. K, Carlson, K. M, Zechiedrich L. 2011. Expression of Multidrug Efflux Pump Genes *acrAB-tolC*, *mdfA*, and *norE* in *Escherichia coli* Clinical Isolates as a Function of Fluoroquinolone and Multidrug Resistance. *Antimicrob Agents Chemother.* **55**:921–914. [PubMed: 21098250].

38. Barrero, M. A. O., Pietralonga, P.A.G, Schwarz, D.G.G., Silva Junior, Paula, A. S.O., Moreira, M.A.S. 2014. Effect of the inhibitor's phenylalanine arginyl s-Naphthylamide (PAsN) and 1-(1-naphthylmethyl)-piperazine (NMP) on expression of genes in multidrug efflux systems of *Escherichia coli* isolates from bovine mastitis. *Research in Veterinary Science*, 97, 176–181.
39. Chaudhary, M., Patnaik, S.K. and Payasi, A. 2014. Evaluation of different drugs in down-regulation of efflux pump genes expression in methicillin-resistant *Staphylococcus aureus* strains. *American Journal of Infectious Diseases*, 10(4), 184.
40. Khan, M.Z.H., Tarek, F.K., Nuzat, M., Momin, M.A. and Hasan, M.R. 2017. Rapid biological synthesis of silver nanoparticles from *Ocimum sanctum* and their characterization. *J. Nanosci.* 2017. 1:1-6.
41. Mohanta, Y.K., Biswas, K., Jena, S.K., Hashem, A., Abd_Allah, E.F. and Mohanta, T.K. 2020. Anti-biofilm and antibacterial activities of silver nanoparticles synthesized by the reducing activity of phytoconstituents present in the Indian medicinal plants. *Front. Microbiol.* <https://doi.org/10.3389/fmicb.2020.01143>.
42. Livak, K.J. and Schmittgen, T.D. 2001. Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. *Methods*. 25(4), 402-408.
43. El-Sayed, M., Algammal, A., Youssef, F. and Hassan, E., 2019., Prevalence and Genetic Characterization of *S. Aureus* Strains Isolated from Raw Milk and Its Products. *Suez Canal Vet. Med. J.* 24(2), 245-256.
44. Bag, M.A.S., Khan, M.S.R., Sami, M.D.H., Begum, F., Islam, M.S., Rahman, M.M., Rahman, M.T. and Hassan, J. 2021. Virulence determinants and antimicrobial resistance of *E. coli* isolated from bovine clinical mastitis in some selected dairy farms of Bangladesh. *Saudi J. Biological Sci.* 28(11), 6317-6323.
45. Brahma, U., Suresh, A., Murthy, S., Bhandari, V. and Sharma, P. 2022. Antibiotic Resistance and Molecular Profiling of the Clinical Isolates of *Staphylococcus aureus* Causing Bovine Mastitis from India. *Microorganisms*. 10(4), 833.
46. Mbindyo, C.M., Gitao, G.C., Plummer, P.J., Kulohoma, B.W., Mulei, C.M. and Bett, R. 2021. Antimicrobial resistance profiles and genes of *Staphylococci* isolated from mastitic cow's milk in Kenya. *Antibiotics*. 10(7), 772.
47. Marami, L.M., Berhanu, G., Tekle, M., Agga, G.E., Beyene, T.J., Tufa, T.B., Beyi, A.F. and Edao, B.M. 2022. Antimicrobial Resistance of *Staphylococci* at Animal Human Interface in Smallholders and Dairy Farms in Central Oromia, Ethiopia. *Infect. Drug Resist.* 15, 3767-3777. doi: 10.2147/IDR.S370592
48. Guo, Y., Song, G., Sun, M., Wang, J. and Wang, Y. 2020. Prevalence and therapies of antibiotic-resistance in *Staphylococcus aureus*. *Front. Cell. Infect. Microbiol.* 10, 107.
49. Halim, R.M.A., Kassem, N.N. and Mahmoud, B.S. 2018. Detection of biofilm producing *Staphylococci* among different clinical isolates and its relation to methicillin susceptibility. Open access. *Macedonian J. Med. Sciences*. 6(8), 1335
50. Vasudevan, P., Nair, M.K.M., Annamalai, T. and Venkitanarayanan, K.S. 2003. Phenotypic and genotypic characterization of bovine mastitis isolates of *Staphylococcus aureus* for biofilm formation. *Vet. Microbiol.* 92(1-2), 179-185
51. Vuotto, C., Longo, F., Pascolini, C., Donelli, G., Balice, M.P., Libori, M.F., Tiracchia, V., Salvia, A. and Varaldo, P.E. 2017. Biofilm formation and antibiotic resistance in *Klebsiella pneumoniae* urinary strains. *J. Appl. Microbiol.* 123(4), 1003-1018
52. ALmiyah, S. 2022. 'Detection of AcrA and AcrB efflux pumps in multidrug-resistant *Klebsiella pneumoniae* that isolated from wounds infection patients in Al-Diwaniyah province'. *Arch. Razi Inst.* doi: 10.22092/ari.2022.358956.2342

53. Annunziato, G. 2019. Strategies to overcome antimicrobial resistance (AMR) making use of non-essential target inhibitors: A review. *Int. J. Molecular Sci.* 20(23), 5844.
54. Maurya N, Jangra M, Tambat R, Nandanwar H. 2019. Alliance of Efflux Pumps with β -Lactamases in Multidrug-Resistant *Klebsiella pneumoniae* Isolates. *Microb Drug Resist.* 25(8), 1155-1163. doi:10.1089/mdr.2018.0414
55. Ramteke, C., Chakrabarti, T., Sarangi, B.K. and Pandey, R.A. 2013. Synthesis of silver nanoparticles from the aqueous extract of leaves of *Ocimum sanctum* for enhanced antibacterial activity. *J. Chemistry*. Article ID 278925 <https://doi.org/10.1155/2013/278925>.
56. Noginov, M.A., Zhu, G., Bahoura, M., Adegoke, J., Small, C., Ritzo, B.A., Draciiiev, V.P and Siialaev, V.M. 2007. The effect of gain and absorption on surface plasmons in metal nanoparticles. *Appl. Phys.* 86 : 455–460.
57. Singhal, G., Bhavesh, R., Kasariya, K., Sharma, A.R. and Singh, R.P. 2011. Biosynthesis of silver nanoparticles using *Ocimum sanctum* (Tulsi) leaf extract and screening its antimicrobial activity. *J. Nanoparticle Res.* 13(7) :2981-2988.
58. Mamidi, G. and Polaki, S.J.A.C. 2019. Synthesis and Characterization of Biogenic silver Nanoparticles and its Antimicrobial Analysis. *J. Appl. Chem*, 8 :112-123.
59. Vanaja, M. and Annadurai, G. 2013. *Coleus aromaticus* leaf extract mediated synthesis of silver nanoparticles and its bactericidal activity. *Appl. Nanosci*, 3(3): 217-223.
60. Siddique, M.H., Aslam, B., Imran, M., Ashraf, A., Nadeem, H., Hayat, S., Khurshid, M., Afzal, M., Malik, I.R., Shahzad, M. and Qureshi, U. 2020. Effect of silver nanoparticles on biofilm formation and EPS production of multidrug-resistant *Klebsiella pneumoniae*. *Biomed. Res. Int.* 2020. ID 6398165 <https://doi.org/10.1155/2020/6398165>.
61. Loo, Y., Y, Rukayadi, Y and Nor-Khaizura, M. A. 2018. In Vitro Antimicrobial activity of Green Synthesized Silver Nanoparticles Against Selected Gram-negative Foodborne Pathogens. *Front Microbiol.* 9:1555.10.3389/fmicb.2018.0155545.
62. Yuan, Y.G., Peng, Q.L. and Gurunathan, S. 2017. Effects of silver nanoparticles on multiple drug-resistant strains of *Staphylococcus aureus* and *Pseudomonas aeruginosa* from mastitis-infected goats: an alternative approach for antimicrobial therapy. *Int. J. Molecular sci.* 18(3) , et569.
62. Kalishwaralal, K., BarathManiKanth, S., Pandian, S.R.K., Deepak, V. and Gurunathan, S. 2010. Silver nanoparticles impede the biofilm formation by *Pseudomonas aeruginosa* and *Staphylococcus epidermidis*. *Colloids. Surfaces Biointerfaces.* 79(2):340-344 7.
63. Sheikh, S., and V. Tale. “Green Synthesis Of Silver Nanoparticles: Its Effect On Quorum Sensing Inhibition Of Urinary Tract Infection Pathogens”. 2017. *Asian J. Pharmaceut. Clin. Res.* 10(5): 302-5, doi:10.22159/ajpcr.2017.v10i5.16949.
64. Gupta, D, Singh, A, Khan , A. U. 2017. Nanoparticles as Efflux Pump and Biofilm Inhibitor to Rejuvenate Bactericidal Effect of Conventional Antibiotics. *Nanoscale Res. Lett.* 12(1):454. doi:10.1186/s11671-017-2222-6.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Green synthesized silver nanoparticles from *Ocimum sanctum*: A potent inhibitor of biofilm forming ability and efflux pumps in bacteria causing bovine mastitis

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Conflicts of interests

The authors declare no competing interests.

Keywords: Antimicrobial resistance, clinical mastitis, biofilm, Green synthesis, silver nanoparticles, efflux pump.

Highlights

- Multidrug resistant (MDR) *Staphylococci*, *Klebsiella pneumoniae* and *Escherichia coli* were isolated from clinical mastitis cases
- The phenotypic and genotypic multidrug resistance profiling was carried out
- Synthesized silver nanoparticles using *Ocimum sanctum* leaf extract as a reducing and capping agent.
- Characterized the AgNPs using UV-vis spectroscopy, FTIR, XRD, SEM -EDAX and TEM
- They exhibited antimicrobial, antibiofilm and efflux pump inhibitory activity in multidrug resistant bacterial isolates
- Green synthesized silver nanoparticles are suitable candidates for combating multidrug resistant bacterial organisms

Dear Editor,

I am writing to submit our manuscript entitled, “Green synthesized silver nanoparticles from *Ocimum sanctum*: A potent inhibitor of multidrug resistance with special reference to biofilm forming ability and efflux pumps in bacteria causing bovine mastitis” for consideration as a research article in the special edition of **Bioinspired nanomaterials: Controlling biofilm and virulence of microbial pathogens** in Journal of Microbial pathogenesis

. This research sought to investigate the phenotypic and genotypic profiling of multidrug resistance in mastitis pathogens and the *invitro* antimicrobial, antibiofilm and efflux pump inhibitory efficacies of green synthesized silver nanoparticles against MDR bacterial isolates. Silver nanoparticles (AgNPs) were biosynthesised using *Ocimum sanctum* leaf extracts and characterized. AgNPs demonstrated antimicrobial, biofilm inhibitory activities and a significant down regulation of expression of the efflux pump genes in the MDR isolates.

Given that therapeutic management of mastitis faces significant challenges due to multidrug resistance and despite advances in understanding mastitis and its challenges, it remains a daunting issue for both veterinarians and livestock owners underscoring a gap in therapeutic arsenal. As antimicrobial resistance escalates, especially in pathogens responsible for clinical mastitis, there is an urgent need to seek innovative therapeutic solutions. While silver nanoparticles have demonstrated promise in various domains of healthcare, their potential against multidrug resistant strains in veterinary medicine, particularly in the context of mastitis remains largely unexplored. This research endeavors to bridge this knowledge gap. This research underscores the potential for development of an ecofriendly alternative in the form of green synthesised silver nanoparticles to combat MDR offering significant antimicrobial, antibiofilm and efflux pump inhibitory activities. The highlighted efflux pump inhibitory function offers a promising avenue for combating antibiotic resistance particularly in the context of bovine mastitis treatment.

This research paper has not been previously published and is not currently under consideration by any other journal. Additionally, all of the authors have approved the contents of this paper and have agreed to the Journal of Microbial pathogenesis submission policies. Each named author has substantially contributed to conducting the underlying research and drafting this manuscript. Additionally, to the best of our knowledge, the named authors have no conflict of interest, financial or otherwise.

Sincerely



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ABSTRACT

Therapeutic management of mastitis faces significant challenges due to multidrug resistance. In the present study multi-drug-resistant (MDR) *Staphylococcus* spp, *Klebsiella pneumoniae* and *Escherichia coli* were isolated from bovine clinical mastitis cases and the phenotypic and genotypic multidrug resistance profiling was carried out. Silver nanoparticles (AgNPs) were biosynthesized using *Ocimum sanctum* leaf extracts and characterized via UV Vis absorption, Fourier Transform Infrared Spectroscopy. Xray Diffraction studies, Energy dispersive spectroscopy and electron Microscopy. The determined minimum inhibitory concentration and minimum bactericidal concentration of the AgNPs against the recovered MDR isolates were 62.5µg/ml and 125 µg/ml respectively. At a concentration of 50 µg/ml, the AgNPs demonstrated biofilm inhibitory activities of 80.35 % for MDR *E. coli*, 71.29 % for *S. aureus* and 60.18 % for MDR *K. pneumoniae*. Post- treatment observations revealed notable differences in biofilm formation across bacterial isolates. Furthermore, AgNP treatment led to significant down regulation of expression of the efflux pump genes *acrB*, *acrE*, *acrF* and *emrB* in Gram- negative isolates and *norB* in *Staphylococci* isolates. This research underscores the potential for the development of an ecofriendly antimicrobial alternative in the form of green synthesized silver nanoparticles to combat drug resistance offering potential antibiofilm and efflux pump inhibitory activities.

Keywords: Antimicrobial resistance, clinical mastitis, biofilm, Green synthesis, silver nanoparticles, efflux pump.

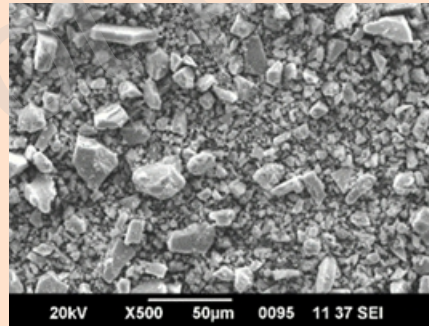
Green synthesized silver nanoparticles from *Ocimum sanctum*: A potent inhibitor of biofilm forming ability and efflux pumps in bacteria causing bovine mastitis



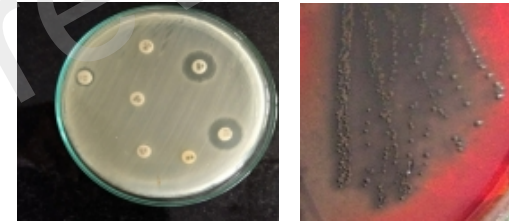
O. sanctum leaves



- XRD
- FTIR
- SEM-EDX
- UV spectra
- TEM



- MDR profiling Biofilm



- MIC - 62.5 µg/ml
- MBC- 125 µg/ml
- Antibiofilm activity
- Efflux pump inhibition

Biosynthesis- AgNPs

Characterization-Ag NPs

Antimicrobial studies

1. Primers used for the amplification of genes for molecular confirmation of the bacterial isolates

S. No	Target gene	Primer sequence 5'-3'	Product amplicon size (bp)	Reference
1	<i>Staph spp</i> 16s rRNA	F-AACTC TGTTATTAGGGAAGAACA R-CC ACCTTCCTCCGGTTTGTACAC	756	McClure <i>et al.</i> (2006)
2	<i>S. aureus</i> <i>nuc</i>	F-TCG CTTGCTATGATTGTG R-GCCAATGTTCTACCATAGC	359	Sasaki <i>et al.</i> (2010)
3	<i>Staphylococcus spp</i> <i>cns</i>	F-TATCCACGAACTTCTAAAACAACCT TTACT R-TCTTTAGATAATACGTATACTTCAGCT TTGAATT	204	Okolie <i>et al.</i> (2015)
4	<i>E. coli</i> <i>uid A</i>	F- AAAACGGCAAGAAAAAGCAG R-ACGCGTGGTTACAGTCTTGCG	147	Bej <i>et al.</i> (1991)
5	<i>K. pneumoniae</i> 16s rRNA	F-CTG CAT TCG AAA CTG GCA GG R-GTT TAC GGC GTG GACT ACCA	187	Singh <i>et al.</i> (2018)

2. The standardized PCR protocol for the amplification of genes for molecular confirmation of the bacterial isolates

S. No	Genes	Denaturation	Annealing	Extension	No. of cycles
1	16s rRNA ¹	94 ⁰ c, 45 s	51 ⁰ c, 90 s	72 ⁰ c ,90 s	30
2	<i>nuc</i> ²	92 ⁰ c, 45 s	49 ⁰ c, 1min	72 ⁰ c, 45 s	35
3	<i>cns</i> ³	92 ⁰ c, 45 s	57 ⁰ c, 1min	72 ⁰ c, 45 s	35
4	<i>E. coli</i> ⁴ , <i>uidA</i>	94 ⁰ c, 40s	55 ⁰ c ,50 s	72 ⁰ c, 50s	35
5	<i>K. pneumoniae</i> <i>16Sr RNA</i>	94 ⁰ c, 45 s	60 ⁰ c, 1min	72 ⁰ c, 1min	35

1. Initial denaturation 95 °c for 2 min, Final extension 72 °c for 10 min
- 2 & 3. Initial denaturation 92 °c for 5 min, Final extension 72 °c for 5 min
4. Initial denaturation 94 °c for 5 min, Final extension 72 °c for 5 min
5. Initial denaturation 94 °c for 5 min, Final extension 72 °c for 10 min

3. Details of primers used for the amplification of genes for antimicrobial resistance in *Staphylococcus* spp

S. No	Target gene	Primer sequence 5'-3'	Product amplicon size (bp)	Reference
1	<i>mec A</i>	F- TGGCTATCGTGTCAATC R- CTGGAACCTTGTGAGCAGAG	310	Alipour <i>et al.</i> (2014)
2	<i>bla Z</i>	F-CAGTTCACATGCCAAAGAG R-TACACTCTTGGCGGTTTC	772	Schnellmann <i>et al.</i> (2006)
3	<i>tet M</i>	F- TTAAATAGTGTCTTGGAG R- CTAAGATATGGCTCTAACAA	656	Ruzaukas <i>et al.</i> (2016)
4	<i>strA/B</i>	F-ATGGTGGACCCTAAACTCT R- CGTCTAGGATCGAGACAAAG	893	Kozak <i>et al.</i> (2009)

4. The PCR protocol for amplification of genes for antimicrobial resistance in *Staphylococcus* spp

S. No	Genes	Denaturation	Annealing	Extension	No. of cycles
1	<i>mec A</i>	92 ⁰ c, 45 s	52 ⁰ c, 1min	72 ⁰ c,45 s	35
2	<i>bla Z</i>	92 ⁰ c, 45 s	50 ⁰ c, 1min	72 ⁰ c,45 s	35
3	<i>tetM</i> -	92 ⁰ c, 45 s	45 ⁰ c,imin	72 ⁰ c,45 s	35
4	<i>strA/str B</i>	940c,30s	630c,1min	720c,30s	35

1,2,3 Initial denaturation 92 °C for 5 min, Final extension 72 °C for 5 min

4. Initial denaturation 94 °C for 10 min, Final extension 72 °C for 10 min

5. Details of primers used in PCR for genes coding for antimicrobial resistance in *E. coli* and *K. pneumoniae*

S. No	Target genes	Primer sequence 5'-3'	Product amplicon size (bp)	References
1	<i>bla</i> SHV	F-GATGAACGCTTTCCCATGATG R-CGCTGTTATCGCTCATGGTAA	214	Yazdi <i>et al.</i> (2012)
2	<i>bla</i> TEM	F-ATGAGTATTCAACATTTCGG R-GTCACAGTTACCAATGCTTA	847	Yazdi <i>et al.</i> (2012)
3	<i>bla</i> CTX-M	F-CGCTTTGCGATGTGCAG R-ACCGCGATATCGTTGGT	550	Ahmed <i>et al.</i> (2004)
4	<i>tet</i> A	F-GCTACATCCTGCTTGCCTTC R-CATAGATCGCCGTGAAGAGG	210	Mazurek <i>et al.</i> (2013)
5	<i>tet</i> B	F-TTGGTTAGGGGCAAGTTTGG R-GTAATGGGCCAATAACACCG	659	Mazurek <i>et al.</i> (2013)
6	<i>tet</i> C	F-CTTGAGAGCCTTCAACCCAG R-ATGGTCGTCATCTACCTGCC	418	Mazurek <i>et al.</i> (2013)
7	<i>strA/str</i> B	F- ATGGTGGACCCTAAACTCT R-CGTCTAGGATCGAGACAAAG	893	Kozak <i>et al.</i> (2009)

6. PCR protocol for amplification of genes coding for antimicrobial resistance in *E. coli* and *K. pneumoniae*

S. No	Genes	Denaturation	Annealing	Extension	No. of cycles
1	<i>bla</i> SHV	94°C, 1min	56°C, 1min	72°C, 1min	35
2	<i>bla</i> TEM	94°C, 1min	56°C, 1min	72°C, 1min	35
3	<i>bla</i> CTX-M	94°C, 1min	56°C, 1min	72°C, 1 min	35
4	<i>tet</i> A	94°C, 1min	60°C, 30s	72°C, 1min	35
5	<i>tet</i> B	94°C, 1min	60°C, 30s	72°C, 1min	35
6	<i>tet</i> C	94°C, 1min	60°C, 30s	72°C, 1min	35
7	<i>strA/str</i> B	94°C, 30s	63°C, 1min	72°C, 30s	35

1,2,3. Initial denaturation 94 °C for 5 min, Final extension 72 °C for 5 min

4,5,6 Initial denaturation 94 °C for 5 min, Final extension 72 °C for 10 min

7. Initial denaturation 94 °C for 10 min, Final extension 72 °C for 10 min

Details of primers used in PCR for amplification of efflux pump genes

S. No	Genes	Primer sequence 5'-3'	Product amplicon size (bp)	Reference
1	<i>acrA</i>	F-CTT AGC CCT AAC AGG ATG G R-TTG AAA TTA CGC TTC AGG T	189	Barreo <i>et al.</i> (2014)
2	<i>acr B</i>	F-CGT ACA CAG AAA GTG CTCA R-CGC TTC AAC TTT GTT TTC TT	183	
3	<i>acr E</i>	F-GCC CTC CTT TAT TCT GAT CT R-GGC TAT ACG ATA AGC ATT G	166	
4	<i>acr F</i>	F-TAG CAA TTT CCT TTG TGG TT R-CCT TTA CCC TCT TTC TCC AT	247	
5	<i>emr A</i>	F-CAA CCG GTA AAG AAG AGG R-GTT ATC GGC CCA GAC TTT G	40	
6	<i>emr B</i>	F-ATT ATG TAT GCC GTC TGC TT R-TTC GCG TAA AGT TAG AGA G	196	Choudhari <i>et al.</i> (2014)
7	<i>norB</i>	F-AGC GCG TTG TCT ATC TTT CC R-GCA GGT GGT CTT GCT GAT AA	213	

PCR conditions for efflux pump genes

S. No	Genes	Denaturation	Annealing	Extension	No. of cycles
1	<i>acrA</i> , <i>acr B</i> , <i>acr E</i> , <i>acr F</i> , <i>emr A</i> , <i>emr B</i>	94°C , 30s	53 °c , 30s	72°C,30s	35
2	<i>norB</i>	94 °C, 15s	51.7 °C,15s	72°C ,15s	35

1. Initial denaturation 94 for 4 min, final extension 72, 5 min
2. Initial denaturation 94 for 4 min, final extension 72, 5 min

Primers for internal control (Swick *et al.*, 2011))

Genes	Sequence
<i>rpsL</i> -F	GCAAAAACGTGGCGTATGTACTC
<i>rpsL</i> -R	TTCGAAACCGTTAGTCAGACGAA

Real time PCR conditions for efflux pump genes

Step	Temperature °C	Time	No. of cycles
Initial denaturation	95	10 min	1
Denaturation	95	15sec	40
Annealing/extension	60	60s	

a. Separate PCR reactions were set up for the target gene and reference (*rpsL*). MDR bacterial isolates were used as samples. Each sample was run in triplicate in a 25 µL reaction. One non-template control for each gene (without template) (with only nuclease-free water) was also included in the reaction.

b. The results were expressed as threshold cycle values (C_T). The relative expression was analyzed using the formula

$$\text{Fold change} = 2^{-\Delta\Delta C_T},$$

where $\Delta\Delta C_T = \Delta C_T \text{ treatment} - \Delta C_T \text{ control}$. (Livak and Schmittgen, 2001)

The results were expressed in fold change as compared to untreated control.



Fig.1.

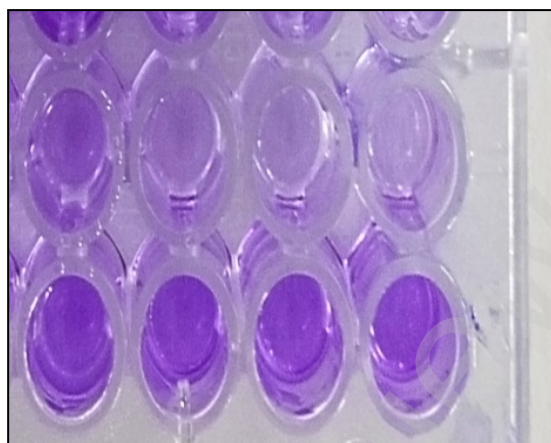


Fig.2.

Fig.1. Formation of black colonies with a dry crystalline consistency indicating strong biofilm production, Fig.2. Crystal violet stained biofilm (A) in the MDR isolates in microtiter plates, Negative control (B).



Fig.3a



Fig.3b

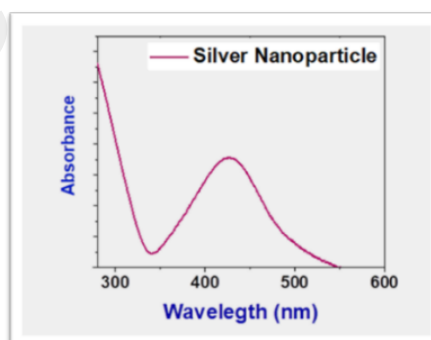


Fig.3.c.

Fig 3a. *Ocimum sanctum* leaves, Fig. 3b. Green synthesized silver nanoparticles, Fig.3c UV Visible spectra of the synthesized silver nanoparticles which depict the absorbance peak centered between 400 and 450 nm indicative of silver nanoparticles.

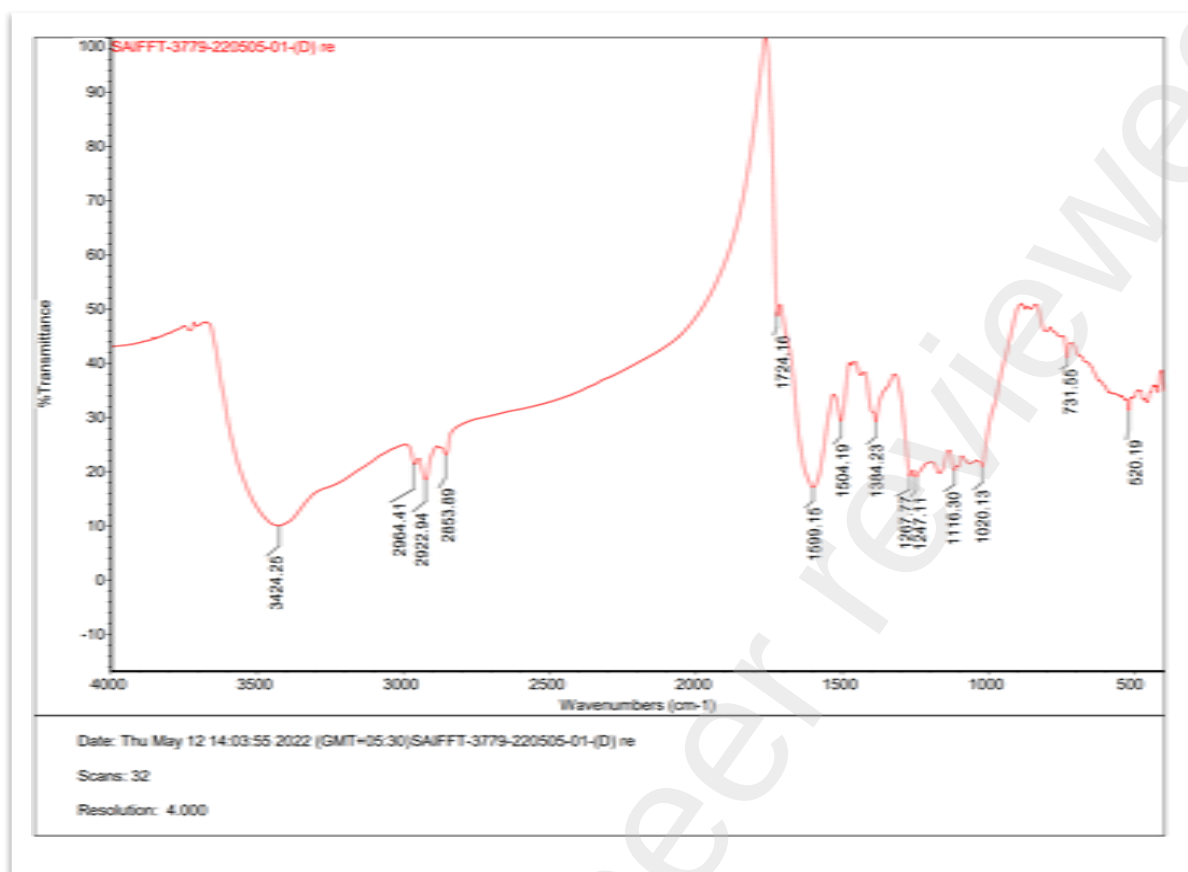


Fig 4. FTIR pattern of biosynthesized silver nanoparticles indicates sharp absorption peaks located at 3434 cm^{-1} , 2964 cm^{-1} , 2922 cm^{-1} , 2853 cm^{-1} , 1724 cm^{-1} , 1599 cm^{-1} which correspond to various functional groups from *O. sanctum* extracts attached to the synthesized silver nanoparticles

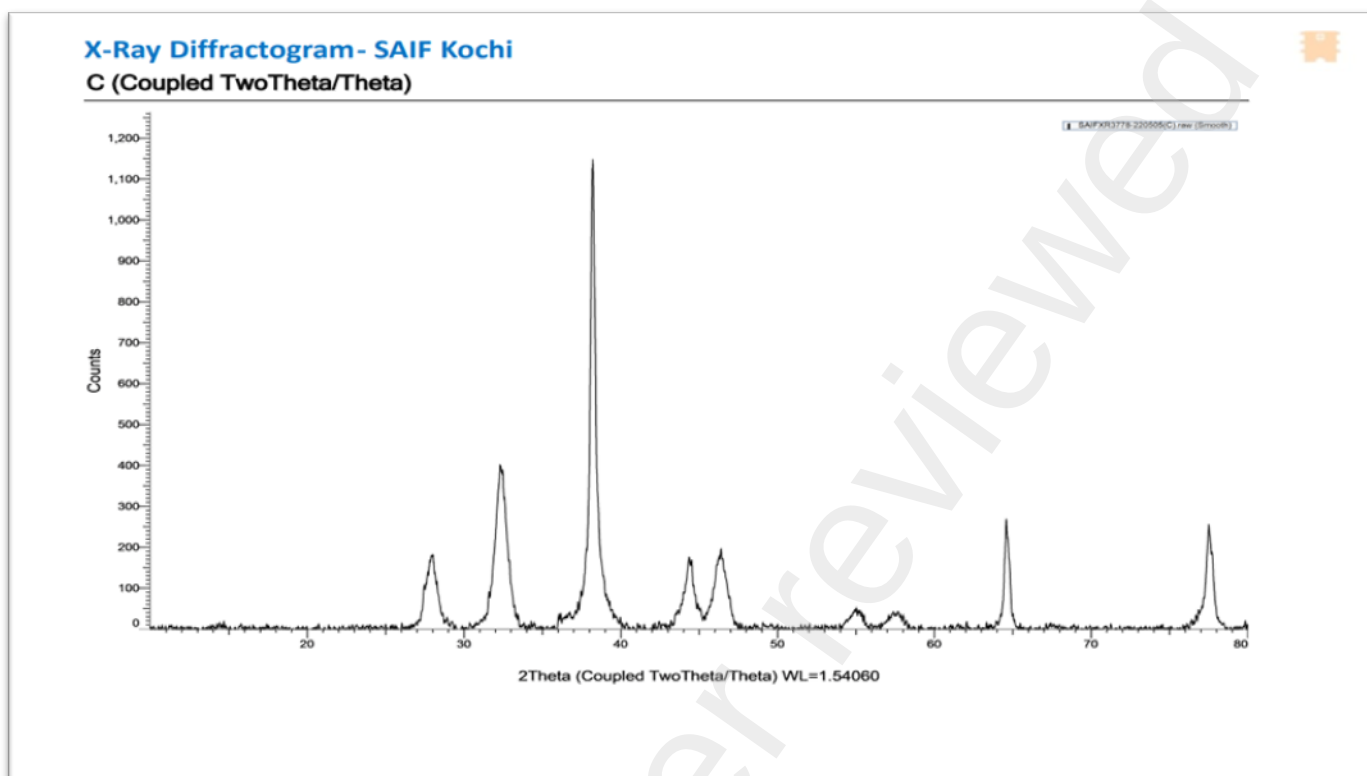


Fig 5. XRD pattern of biosynthesised silver nanoparticles indicates distinct peaks at 77.513 °, 64.596 °, 46.393 °, 44.408 °, 38.211 °, 36.170 °, 32.336 ° and 27.954 ° characteristic of silver nanoparticles

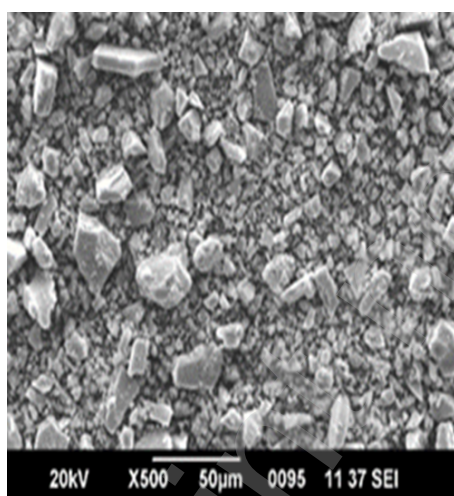


Fig.6.

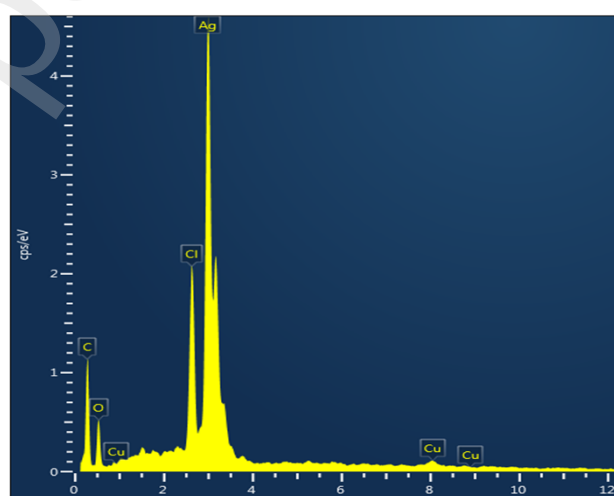


Fig.7.

Fig 6. SEM images of the silver nanoparticles indicates the crystalline nature, shape of the crystals and the even distribution.

Fig 7. EDAX quantitative analysis registered a predominance of elemental silver, accounting for approximately 58.75 (wt. %)

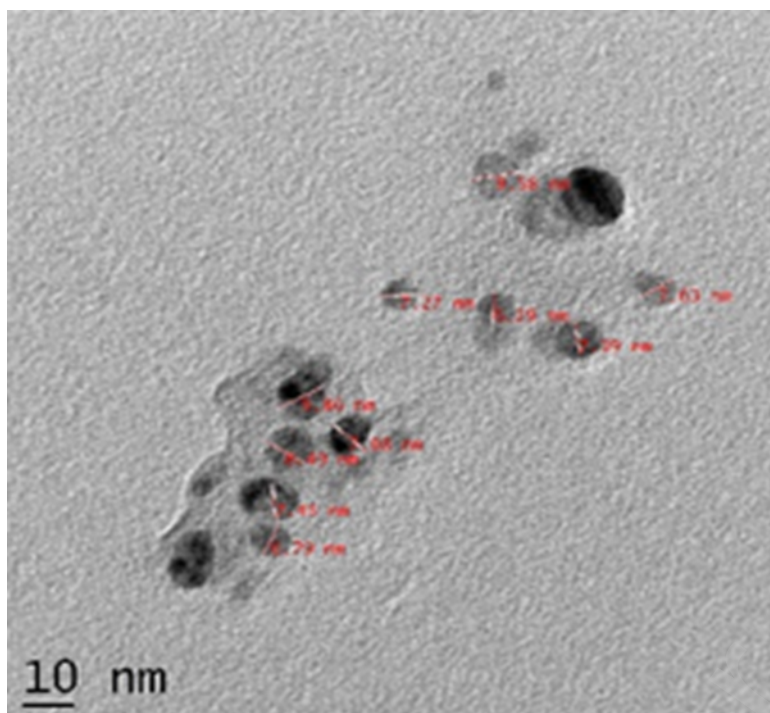


Fig 8. TEM image of the silver nanoparticles indicating Ag nanoparticles were spherical in shape with particles in size range within 6.29-10.61 nm with mean diameter $\pm$ SD of 8.287 ± 1.389 nm

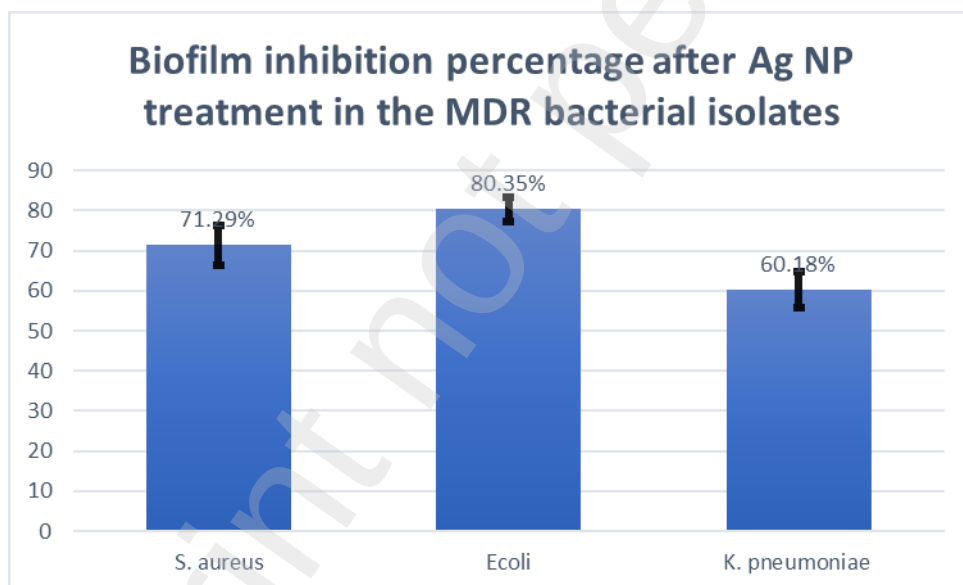


Fig 9. X axis-Biofilm inhibition percentage after Ag NP treatment in the MDR bacterial isolates. *S. aureus* - 71.29 % drop in the biofilm production, *E. coli* - 80.35 % drop in the biofilm production, *K. pneumoniae*- 60.18 % drop in the biofilm production. Y axis- Biofilm inhibition percentage

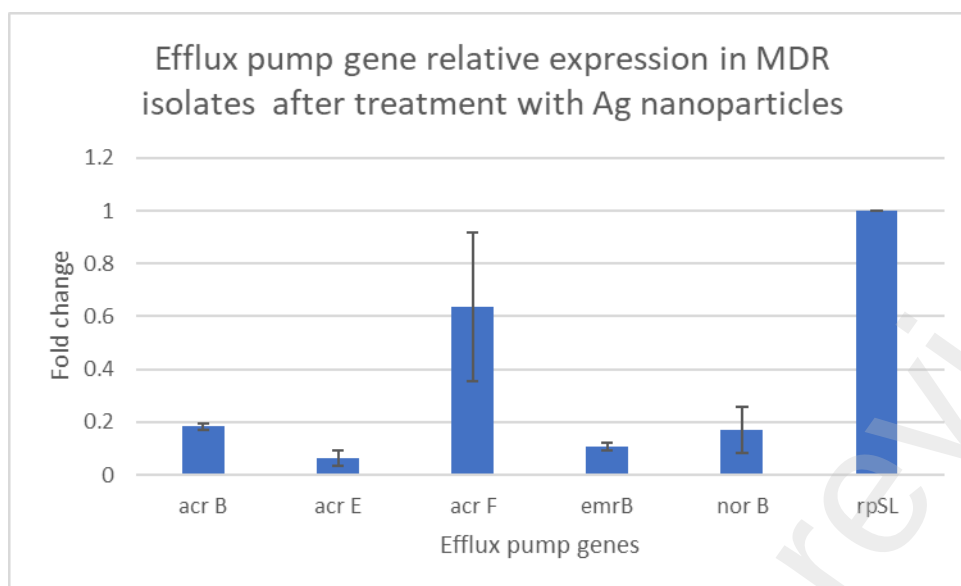


Fig.10. Efflux pump gene relative expression in MDR strains is represented as a fold change between the efflux pump genes and the internal control gene *rpSL* after treatment with sub MIC concentration of Ag NPs. Data are mean $\pm$ SE of three independent experiments.

Table.1. Multidrug resistance profile of the MDR isolates

Isolate Id. No	Antibiotics							MDR pattern to different classes of antibiotics	MAR index	
	TET	ENR	GEN	β lactam antibiotics						SUT
				CPZ	CFT-TZ	AMXCLV	MET			
SA 07	R	R	R		R	R	R	R	TET+ENR+GN+BL + SUT (5)	0.875
SS04	R	R			R				TET+ENR+BL (3)	0.375
EC 04	R	R	R	R	R				TET+ENR+GEN+BL (4)	0.571
KN07	R		R		R	R			TET+GEN+BL (3)	0.571

TET- Tetracycline ENR- Enrofloxacin BL- β lactams SUT- Sulpha -trimethoprim GN- Gentamicin CPZ- Cefoperazone CFT-TZ – Ceftriaxone- Tazobactam AMXCLV- Amoxicillin - Clavulanate
 SA01- *S. aureus*, SS01-*S. saprophyticus*, EC-*E. coli* KN -*K. pneumoniae* R-Resistant

Table 2. Genotypic resistance profile and biofilm formation of the MDR isolates

Isolate ID	Biofilm formation		Gene pattern	
	CRA	MTP OD $\pm$ SE	ARGs	Efflux pump genes
SA 07	Strong biofilm producer	4xOD _c < OD 0.21 $\pm$ 0.02	<i>mecA</i> , <i>strA/strB</i>	-
SS04	Strong biofilm producer	4xOD _c < OD 0.21 $\pm$ 0.006	<i>mecA</i> , <i>blaZ</i> , <i>strA/strB</i>	<i>norB</i>
EC 04	Strong biofilm producer	4xOD _c < OD 0.324 $\pm$ 0.01	<i>bla</i> TEM,	<i>acrB</i> , <i>acrE</i> , <i>acrF</i> , <i>emrB</i>
KN07	Strong biofilm producer	2OD _c < OD $\leq$ 4xOD _c 0.144 $\pm$ 0.006	<i>bla</i> CTX-M, <i>tetA</i> , <i>tetB</i>	<i>acrB</i> , <i>acrE</i> , <i>acrF</i> , <i>emrB</i>

SA01- *S. aureus*, SS01-*S. saprophyticus*, EC04-*E. coli*, KN 07 -*K. pneumoniae*, CRA- Congo red test, MTP-Microtitre Plate assay, ARGs- Antibiotic resistance genes, OD-Optical density, OD_c . Optical density of the control well- (0.052)

Table. 3. Relative expression of the efflux pump genes in MDR isolates

Genes	ΔC_T control	ΔC_T treatment	$\Delta \Delta C_T$ (ΔC_T treatment – ΔC_T control)	Fold of expression (Treatment)
<i>acr B</i>	4.73 $\pm$ 0.101	7.19 $\pm$ 0.003	2.47 $\pm$ 0.10 ^{ab}	0.183 $\pm$ 0.013

<i>acr E</i>	5.33 ± 0.395	9.80 ± 0.546	4.47 ± 0.71 ^a	0.064 ± 0.029
<i>acrF</i>	5.90 ± 0.808	7.18 ± 0.048	1.29 ± 0.84 ^b	0.637 ± 0.282
<i>emrB</i>	1.72 ± 0.227	5.02 ± 0.064	3.26 ± 0.20 ^{ab}	0.106 ± 0.014
<i>nor B</i>	4.37 ± 0.442	7.37 ± 0.700	3.00 ± 0.82 ^{ab}	0.170 ± 0.089
F-value (P-value)			3.683* (0.032)	

* Significant at 0.05 level; ns non-significant

Means having different letter as superscript differ significantly within a column. Significant F-value between treated and untreated shows that there exists significant difference in ΔC_T values between treated and untreated groups.